Supplementary material

Per-joint position verdicts between markerless triangulation variants depend on the joint-centre offset they contain

Every table below is reproduced from the result file named in its heading; no number has been retyped. Effects are signed so that a positive value means the second-named arm has the lower error. The result files also carry annotations written during analysis. The rules recording what a result would be taken to mean before it was seen are kept, headed HOW TO READ THIS, because a reader needs them to judge the result. Working notes addressed to the authors -- reminders, cross-checks, and instructions about which analysis to prefer -- are omitted here and are present in the files distributed with the code.

A note on angle names

The manuscript names its angle outcomes for the segments that define them, because they are three-point included angles and not model-based joint kinematics: the trunk-thigh angle moves with trunk lean and pelvic tilt, and the frontal-plane form is a projection rather than an ab/adduction angle under any standard convention. The result files below were written before that renaming and print the original keys. They map one to one:

result files	manuscript	approximates
`knee_flex`	thigh-shank angle	knee flexion
`knee_abd`	frontal-plane thigh-shank angle	knee ab/adduction
`hip_flex`	trunk-thigh angle	hip flexion
`dflex_toe`, `dflex_heel`	shank-foot angle	ankle dorsiflexion

No number changes with the name; only the label does.

Table S0. Rig, reference and software

Reproducibility detail moved out of Section 2.1 so the main text carries only what the argument uses.

item	**value**
video cameras	9 machine-vision, 1920x1080, 200 Hz
ring geometry	median radius 4.46 m; 3.0-6.1 m from the centroid
focal lengths	non-uniform, a 1.34x spread across the ring
distortion	substantial radial distortion in all 99 calibrations
reference system	15-camera Qualisys Oqus, 200 Hz
reference model	the dataset's own 6-DoF full-body model, Visual3D v6
reference filter	Butterworth, 4th order, 12 Hz
joint centres	the archive record states that joint centres are the midpoint of the medial and lateral markers for every joint except the hip, which uses the regression equations of Bell et al. (1989). The knee and ankle references are therefore direct marker constructions; the hip reference is a prediction from external landmarks, whose own error is of a magnitude comparable to the offset reported here and is mirrored across the midline like an anatomical offset. The hip result therefore cannot be apportioned between the two systems, and that is now a statement about a known construction rather than an unknown one.
frames per trial	60, at near-even spacing across the trial (roughly every ninth), a compute bound on re-running every arm over every frame; the spacing spans the whole pass through the capture volume rather than a window of it.
2D detector	RTMPose-m via `rtmlib` 0.0.15, ONNX Runtime 1.27.0
undistortion	`cv2.undistortPoints`, OpenCV 5.0.0

Table S1. The complete contrast family (*m* = 50)

As in Table S3n, the block below is verbatim and still prints q-values on its leave-one-out offset-table rows. Those are withheld by the rule of Section 2.5; the manuscript's counts exclude them.

The main tables display the contrasts the argument turns on. This is the whole family, unfiltered and in the order Benjamini–Hochberg sees it, so that the correction can be checked rather than taken on trust. `[pos:AW]` denotes ankles and wrists; `[pos:HKA]` the hip, knee and ankle from which the joint angles are built. *k* is the number of participants contributing; all contrasts use the exact sign-flip test over 2^{11} = 2048 patterns, so the smallest attainable p is 0.000977, and contrasts reported at that floor cannot be ranked against one another.

Source: `BIOCV_PERM_V3.txt` (`biocv_permutation_v2.py`).

Two arm orders appear in this block. The [pos:AW] rows name the no-rejection arm first ("no rejection vs reprojection rejection", +0.131 mm) and the [pos:HKA] rows name the rejection arm first ("reproj rejection vs none", +0.005 mm), because the two levels were accumulated in different passes. Every label states its own order and every sign follows it, so the two are consistent; but a reader comparing rows across levels must read the label before the sign. Table 1(a) prints the [pos:HKA] contrast in the first order, as −0.005 mm. Regenerating this file to unify the orders costs 3.3 hours of compute and would move no number, so the convention is stated rather than rewritten.

```
EXACT SIGN-FLIP PERMUTATION TEST + BENJAMINI-HOCHBERG FDR
Authoritative contrast family for the manuscript.
11 subjects -> 2^11 = 2048 sign patterns, fully enumerated.
No asymptotics, no coverage assumption. Minimum attainable two-sided p = 0.000977.
Family size m = 50 contrasts. POSITIVE effect = the SECOND-NAMED arm is better
(i.e. effect = mean|error| of the first arm minus mean|error| of the second).
EVERY label names its two arms in the SAME order they are subtracted.

LEVELS:  [pos:AW]  3D position over ankles + wrists (the original joint set)
         [pos:HKA] 3D position over hip/knee/ankle -- the SAME joints the angles
                   are built from, same frames, same candidate cameras, same arms.
         knee_flex / knee_abd / hip_flex  joint-angle outcomes

contrast                                                     n   effect   exact p     BH q  verdict
[pos:AW] uniform vs confidence weighting                    11   +5.062    0.0010   0.0035  SURVIVES FDR
[pos:AW] 1/d alone vs confidence                            11   +4.971    0.0010   0.0035  SURVIVES FDR
[pos:AW] does 1/d refine confidence?                        11   +0.256    0.0205   0.0380  SURVIVES FDR
[pos:AW] reprojection vs object-space criterion             11   +0.332    0.0010   0.0035  SURVIVES FDR
[pos:AW] reproj rejection vs the same + 1/d weighting       11   +0.780    0.0010   0.0035  SURVIVES FDR
[pos:AW] object-space rejection vs reproj+1/d weighting     11   +0.448    0.0010   0.0035  SURVIVES FDR
[pos:AW] no rejection vs reprojection rejection             11   +0.131    0.5352   0.6223  n.s.
[pos:AW] no rejection vs object-space rejection             11   +0.463    0.0664   0.1036  n.s.
[pos:AW] criterion, both 1/d-weighted                       11   -0.072    0.0391   0.0651  raw only
[pos:AW] matched-k=1: reprojection vs object-space at equal retention  11   +0.148    0.0029   0.0086  SURVIVES FDR
[pos:AW] matched-k=2: reprojection vs object-space at equal retention  11   +0.428    0.0010   0.0035  SURVIVES FDR
[pos:AW] matched-k=3: reprojection vs object-space at equal retention  11   +0.782    0.0010   0.0035  SURVIVES FDR
[pos:HKA] uniform vs confidence weighting                   11   +0.917    0.0010   0.0035  SURVIVES FDR
[pos:HKA] 1/d weighting, no rejection                       11   -0.691    0.0010   0.0035  SURVIVES FDR
[pos:HKA] reproj rejection vs none                          11   +0.005    0.9639   0.9639  n.s.
[pos:HKA] objd rejection vs none                            11   -0.163    0.1416   0.1914  n.s.
[pos:HKA] reprojection vs object-space criterion            11   +0.168    0.0078   0.0178  SURVIVES FDR
[pos:HKA] matched-k=1: reprojection vs object-space at equal retention  11   +0.029    0.6152   0.6687  n.s.
[pos:HKA] matched-k=2: reprojection vs object-space at equal retention  11   +0.039    0.5693   0.6467  n.s.
[pos:HKA] matched-k=3: reprojection vs object-space at equal retention  11   +0.169    0.1191   0.1655  n.s.
[pos:HKA] LOO population offset correction                  11   +9.848    0.0010   0.0035  SURVIVES FDR
[pos:HKA] GN: no rejection vs reprojection rejection        11   +0.858    0.0020   0.0061  SURVIVES FDR
```

[pos:HKA] GN: reprojection vs object-space criterion	11	-0.124	0.0801	0.1178	n.s.
[pos:HKA] DLT vs GN, no rejection	11	-1.384	0.0010	0.0035	SURVIVES FDR
knee flex: uniform vs confidence weighting	11	+0.023	0.5820	0.6467	n.s.
frontal knee: uniform vs confidence weighting	11	+0.078	0.0010	0.0035	SURVIVES FDR
hip flex: uniform vs confidence weighting	11	+0.227	0.0020	0.0061	SURVIVES FDR
knee flex: reproj rejection vs none	11	+0.188	0.0098	0.0212	SURVIVES FDR
knee flex: objd rejection vs none	11	+0.141	0.0039	0.0109	SURVIVES FDR
knee flex: reprojection vs object-space criterion	11	+0.047	0.1494	0.1966	n.s.
knee flex: 1/d weighting, no rejection	11	+0.052	0.3574	0.4468	n.s.
knee flex: rejection under best weighting	11	+0.114	0.0391	0.0651	raw only
frontal knee: reproj rejection vs none	11	+0.013	0.7266	0.7619	n.s.
frontal knee: objd rejection vs none	11	+0.067	0.0352	0.0628	raw only
frontal knee: reprojection vs object-space criterion	11	-0.054	0.0684	0.1036	n.s.
frontal knee: 1/d weighting, no rejection	11	-0.188	0.0010	0.0035	SURVIVES FDR
hip flex: reproj rejection vs none	11	+0.192	0.0107	0.0224	SURVIVES FDR
hip flex: objd rejection vs none	11	+0.167	0.0146	0.0282	SURVIVES FDR
hip flex: reprojection vs object-space criterion	11	+0.025	0.5107	0.6080	n.s.
hip flex: 1/d weighting, no rejection	11	-0.219	0.0010	0.0035	SURVIVES FDR
knee flex matched-k=1: reprojection vs object-space at equal retention	11	-0.089	0.0068	0.0163	SURVIVES FDR
knee flex matched-k=2: reprojection vs object-space at equal retention	11	-0.129	0.0049	0.0128	SURVIVES FDR
knee flex matched-k=3: reprojection vs object-space at equal retention	11	-0.051	0.3389	0.4344	n.s.
knee flex: LOO population offset correction	11	-0.606	0.0068	0.0163	SURVIVES FDR
frontal knee: LOO population offset correction	11	-0.149	0.4434	0.5407	n.s.
hip flex: LOO population offset correction	11	+0.164	0.7314	0.7619	n.s.
knee flex GN: no rejection vs reprojection rejection	11	-0.169	0.0146	0.0282	SURVIVES FDR
knee flex GN: reprojection vs object-space criterion	11	+0.045	0.1045	0.1493	n.s.
knee flex: DLT vs GN, no rejection	11	-0.004	0.9600	0.9639	n.s.
knee flex: plain DLT vs best-GN	11	-0.173	0.0674	0.1036	n.s.

Table S2. Distortion characterisation

The two criteria are distinguished by near-camera residuals (Eq. 1), which is where a radial-distortion model is least constrained, so the criterion comparison is only interpretable if the distortion behaviour is characterised. S2a fits the exponent α in detector pixel error ∝ d^α and asks whether it reflects genuine heteroscedasticity or the joint-centre offset seen in perspective; S2b repeats the principal contrasts on undistorted image coordinates.

S2b should be read in both directions. Correcting distortion removes the object-space advantage in the highest camera-asymmetry bin; it also raises the pooled advantage and turns the two lowest bins positive. The criterion comparison at `[pos:AW]` is therefore sensitive to the distortion model rather than independent of it, which is the limitation Section 3.2 states.

S2a. The distance–error exponent — `BIOCV_ALPHA.txt`

```
IS THE DISTANCE DEPENDENCE REAL HETEROSCEDASTICITY, OR THE JOINT-CENTRE OFFSET
SEEN IN PERSPECTIVE?  (BIOCV_LM_NOISE_BIG.txt reported alpha = -1.111 on a wider
joint set; this file recomputes it on the ten joints listed below -- shoulders,
wrists, hips, knees and ankles, NOT the lower limb -- hence the small difference.)
N=556300 camera-joint observations, 11 subjects, joints [5, 6, 9, 10, 11, 12, 13, 14, 15, 16].
Reference for 'pixel error' differs per panel; everything else is identical.

--- (1) vs projection of VICON GT  [this is what BIOCV_LM_NOISE_BIG.txt did] ---
       dist bin (mm)       n  median e_px  mean e_px  p90 e_px
           2096-3582  111260        11.36      13.05     22.29
           3582-4005  111260         9.88      12.02     20.71
           4005-4712  111260         8.07       9.74     17.12
           4712-5681  111260         6.64       7.39     13.26
           5681-8764  111260         5.35       5.92     10.64
power-law fit on medians: sigma_px ~ d^alpha with alpha = -1.184

--- (2) vs projection of GT + LEAVE-ONE-SUBJECT-OUT mean offset ---
       dist bin (mm)       n  median e_px  mean e_px  p90 e_px
           2096-3582  111260         7.24       9.75     18.67
           3582-4005  111260         6.85       9.69     18.17
           4005-4712  111260         5.55       7.60     13.90
           4712-5681  111260         4.41       5.37     10.07
           5681-8764  111260         3.47       4.19      7.93
power-law fit on medians: sigma_px ~ d^alpha with alpha = -1.200

--- (3) vs projection of GT + THAT SUBJECT'S OWN mean offset (in-sample bound) ---
       dist bin (mm)       n  median e_px  mean e_px  p90 e_px
           2096-3582  111260         6.13       8.54     16.49
           3582-4005  111260         5.70       8.60     16.66
           4005-4712  111260         4.66       6.71     12.32
           4712-5681  111260         3.71       4.59      8.55
           5681-8764  111260         2.83       3.51      6.59
power-law fit on medians: sigma_px ~ d^alpha with alpha = -1.242

=== VERDICT ===
alpha(raw GT)      = -1.184
alpha(LOO-offset)  = -1.200
alpha(in-sample)   = -1.242

|alpha| GROWS by 5% once the joint-centre offset is removed.
```

S2b. Undistortion control — `BIOCV_UNDISTORT.txt`

```
DECISIVE TEST: does uncorrected lens distortion manufacture the asymmetry effect?
N=24960 joint-observations, 11 subjects. Distortion applied via cv2.undistortPoints.

--- (1) DIAGNOSTIC: distortion displacement at the actual detections ---
distortion displacement (px): mean=6.09 median=1.62 p90=18.41 max=84.80
  (compare: effect sizes are ~0.05 px equivalent for the pooled +0.15mm, ~1.5-2 px for +4.6mm)
corr(distortion displacement, camera-to-joint distance) = -0.235
  NEAR cameras (bottom quartile dist, 3364mm): mean distortion 8.33 px, mean image radius 365 px
  FAR  cameras (top quartile dist, 6300mm): mean distortion 2.95 px, mean image radius 246 px
  -> the near-camera mechanism requires NEAR >> FAR distortion. Ratio near/far = 2.82

--- (2a) UNCORRECTED (pinhole projection, distorted detections) ---
  asym bin       n  mean_re  mean_di   dMEAN         dMEAN 95%CI
      <1.5       0   few
     1.5-2   13239     16.5     16.7   -0.13      [-0.38,+0.07]
     2-2.5    8203     34.9     35.0   -0.13      [-0.30,+0.07]
     2.5-3    3023     63.1     61.8   +1.35      [+0.61,+1.98]*
       3-4     495     82.4     77.8   +4.62      [+2.16,+8.45]*
        >4       0   few
POOLED: dMEAN=+0.147 mm, CI [+0.036,+0.257]  | mean abs error reproj=29.5 objd=29.4 mm

--- (2b) DISTORTION-CORRECTED (detections undistorted to ideal pinhole) ---
  asym bin       n  mean_re  mean_di   dMEAN         dMEAN 95%CI
      <1.5       0   few
     1.5-2   13239     15.9     15.7   +0.16      [+0.08,+0.26]*
     2-2.5    8203     16.8     16.3   +0.47      [+0.34,+0.61]*
     2.5-3    3023     17.3     16.7   +0.67      [+0.43,+1.00]*
       3-4     495     20.4     19.7   +0.68      [-1.39,+2.23]
        >4       0   few
POOLED: dMEAN=+0.334 mm, CI [+0.225,+0.439]  | mean abs error reproj=16.5 objd=16.1 mm
```

S2c. Weighting-factor sweep — `BIOCV_WEIGHTFACTOR.txt`

This file reports the ankles-and-wrists confidence-versus-c/d effect as +0.246 mm against +0.256 in Table S1 and Table 1(a). The gap is the pooled-versus-per-participant difference set out under Table 3(a): this sweep pools observations, while the contrast families average per-participant effects, which is the inferential unit. Where the two differ the per-participant value is the tested one.

The weighting arms of Section 2.3 compared against confidence weighting on both position and knee angle, including the c/d^2 variant, which is null on both.

```
WEIGHT FACTORISATION -- which factor in the DLT weights is doing the work?
11 subjects. NO rejection in any arm; only weighting differs.

--- DIAGNOSTIC: is detector confidence itself distance-dependent? ---
corr(confidence, camera distance) = +0.145   (n=222520)
   distance bin (mm)        n  mean conf  median conf
           2096-3737    55630      0.813        0.838
           3737-4331    55630      0.796        0.846
           4331-5442    55630      0.835        0.869
           5442-8764    55630      0.852        0.874
power-law fit: confidence ~ d^+0.093
  If beta is strongly negative, 'conf' already encodes distance and dividing by d
  again double-counts it -- the same error as combining distance rejection + weighting.

--- 3D POSITION error, peripheral joints (N=24960 obs) ---
   weighting      mean          vs conf (95% CI)
     uniform    21.678   -5.086 [-5.844,-4.428]*
        conf    16.592               (reference)
       inv_d    21.604   -5.012 [-5.879,-4.295]*
      conf_d    16.346   +0.246 [+0.073,+0.427]*
     conf_d2    16.591   +0.001 [-0.326,+0.356]
   (mm; POSITIVE = better than the current confidence weighting)

--- KNEE ANGLE error (N=12480 obs) ---
   weighting      mean          vs conf (95% CI)
     uniform     2.607   -0.032 [-0.099,+0.040]
        conf     2.575               (reference)
       inv_d     2.586   -0.011 [-0.168,+0.141]
      conf_d     2.528   +0.047 [-0.057,+0.145]
     conf_d2     2.550   +0.025 [-0.164,+0.195]
   (deg; POSITIVE = better than the current confidence weighting)
```

S2d. The same distortion control at [pos:HKA] — `BIOCV_UNDISTORT_HKA.txt`

S2b ran at [pos:AW] only, which left every hip/knee/ankle criterion result conditional on a distortion model no test had exercised at that joint set. This is the same control on the joint set the paper's main claims are made on. It behaves the same way: uncorrected, the object-space advantage concentrates in a high-asymmetry bin (+2.49 mm at 2.5-3, CI excluding zero) and the pooled figure is +0.222 mm; corrected, the bin structure flattens to +0.10 to +0.26 mm across every bin and the pooled figure is +0.164 mm. The asymmetry concentration is therefore a distortion artefact at [pos:HKA] as it is at [pos:AW], and the small pooled advantage survives correction at both. Note that correction also lowers the mean absolute position error from 38.6 to 28.5 mm, an order of magnitude more than any effect contrasted in this paper, which is why the correction is applied throughout rather than treated as a sensitivity. Intervals are percentile bootstrap over PARTICIPANTS, B = 3000; where a bootstrap interval and the exact sign-flip test disagree, Section 2.5 gives the exact test.

```
DECISIVE TEST: does uncorrected lens distortion manufacture the asymmetry effect?
N=37440 joint-observations, 11 subjects, joint set [pos:HKA]. Distortion applied via cv2.undistortPoints.
All intervals below are percentile bootstrap, B=3000, RESAMPLING SUBJECTS (not observations), so they carry the within-subject clustering.

--- (1) DIAGNOSTIC: distortion displacement at the actual detections ---
distortion displacement (px): mean=5.76 median=1.37 p90=17.80 max=84.80
  (compare: effect sizes are ~0.05 px equivalent for the pooled +0.15mm, ~1.5-2 px for +4.6mm)
corr(distortion displacement, camera-to-joint distance) = -0.232
  NEAR cameras (bottom quartile dist, 3386mm): mean distortion 7.73 px, mean image radius 357 px
  FAR  cameras (top quartile dist, 6288mm): mean distortion 2.84 px, mean image radius 239 px
  -> the near-camera mechanism requires NEAR >> FAR distortion. Ratio near/far = 2.72

--- (2a) UNCORRECTED (pinhole projection, distorted detections) ---
  asym bin       n  mean_re  mean_di   dMEAN         dMEAN 95%CI
      <1.5       0   few
     1.5-2   20621     27.6     27.7   -0.12      [-0.39,+0.13]
     2-2.5   11644     43.5     43.6   -0.15      [-0.65,+0.38]
     2.5-3    4831     70.0     67.5   +2.49      [+0.88,+4.18]*
       3-4     344     96.1     94.9   +1.22      [-1.29,+3.32]
        >4       0   few
POOLED: dMEAN=+0.222 mm, CI [+0.039,+0.399]  | mean abs error reproj=38.6 objd=38.4 mm

--- (2b) DISTORTION-CORRECTED (detections undistorted to ideal pinhole) ---
  asym bin       n  mean_re  mean_di   dMEAN         dMEAN 95%CI
      <1.5       0   few
     1.5-2   20621     27.5     27.4   +0.10      [-0.00,+0.19]
     2-2.5   11644     29.4     29.2   +0.26      [+0.08,+0.42]*
     2.5-3    4831     29.9     29.7   +0.21      [-0.19,+0.65]
       3-4     344     34.3     34.0   +0.26      [-2.63,+3.22]
        >4       0   few
POOLED: dMEAN=+0.164 mm, CI [+0.079,+0.254]  | mean abs error reproj=28.5 objd=28.3 mm
```

Table S3. Mechanism analyses

These are the analyses behind the account of *why* a criterion that helps position can harm an angle: the effect is not uniform across joints, and the joints it helps and harms sit on the same segment.

S3a. Decomposition by camera-distance asymmetry — `BIOCV_ASYM_HKA.txt`

Position error at the hip, knee and ankle binned by how asymmetric the contributing cameras' distances are, which is the regime Eq. 1 says separates the two criteria. The per-joint decomposition the main text uses is S3b.

```
DOES REJECTION'S HIGH-ASYMMETRY POSITION BENEFIT SURVIVE AT [pos:HKA]?
N=37440 joint-observations over hip/knee/ankle, 11 subjects.
Compare with BIOCV_WEIGHTING.txt, which ran the same contrast on ankles+WRISTS.
POSITIVE = the rejection arm is better (lower error) than no rejection.

  asym bin       n   none mm  reproj mm   objd mm  keepRe  keepOb  none-rep       p  none-objd       p
   1.5-2.0   20621     27.41      27.49     27.39    8.48    8.56     -0.08  0.4629      +0.00  0.9619
   2.0-2.5   11644     29.46      29.44     29.18    8.19    8.45     -0.03  0.8369      +0.25  0.0215
   2.5-3.0    4831     30.17      29.92     29.70    7.59    8.01     +0.12  0.7344      +0.42  0.1104
   3.0-4.0     344     37.17      34.27     34.01    7.13    7.60     +4.32  0.6133      +5.33  0.1367
       ALL   37440     28.49      28.47     28.31    8.26    8.45     -0.01  0.9639      +0.16  0.1416

Share of hip/knee/ankle observations at asymmetry >= 3: 0.92%  (n=344)
```

S3b. Per-joint family with BH correction — `BIOCV_PERJOINT_FDR.txt`

The twelve per-joint tests with their corrected q-values. Table 1(b) of the main text prints the effects and unadjusted p; the correction is here. The family is four interventions crossed with three lower-limb joints — the four for which a per-joint decomposition was computed — and the pooled `[pos:HKA]` rows are excluded because they belong to the $m = 50$ family.

```
BENJAMINI-HOCHBERG OVER THE PER-JOINT FAMILY
Family size m = 12: four interventions x three lower-limb joints. Exact sign-flip
p-values taken verbatim from BIOCV_PERJOINT_GN.txt (Part 1). The pooled [pos:HKA] rows are excluded
because they belong to the m=50 family; including them would double-count.

POSITIVE effect = the second-named arm has the lower error at that joint.

intervention                  joint  baseline  effect mm   rel %        p     BH q  verdict
uniform -> confidence           hip     39.31     +1.564   +3.98   0.0010   0.0040  SURVIVES
uniform -> confidence          knee     30.03     +0.365   +1.22   0.0010   0.0040  SURVIVES
uniform -> confidence         ankle     15.50     +0.822   +5.30   0.0010   0.0040  SURVIVES
adaptive criterion             knee     30.58     +0.339   +1.11   0.0020   0.0040  SURVIVES
criterion at matched k=2        hip     38.12     +0.530   +1.39   0.0020   0.0040  SURVIVES
criterion at matched k=2       knee     32.32     -0.655   -2.03   0.0020   0.0040  SURVIVES
criterion at matched k=1       knee     31.33     -0.322   -1.03   0.0029   0.0050  SURVIVES
adaptive criterion            ankle     15.07     +0.190   +1.26   0.0352   0.0528  raw only
criterion at matched k=2      ankle     15.56     +0.219   +1.40   0.0762   0.1008  n.s.
criterion at matched k=1        hip     37.74     +0.253   +0.67   0.0840   0.1008  n.s.
criterion at matched k=1      ankle     15.18     +0.134   +0.88   0.1484   0.1619  n.s.
adaptive criterion              hip     38.70     -0.024   -0.06   0.8350   0.8350  n.s.

7 of 12 survive BH; 8 were significant unadjusted.
Contrasts significant unadjusted that do not survive correction are exploratory.
```

S3c. Posture dependence — `BIOCV_POSTURE.txt`

```
IS THE JOINT-CENTRE OFFSET POSTURE-DEPENDENT?
N=37440 joint-observations, 11 participants.
Arm = confidence weights, no rejection. Offsets are (estimate - ground truth) in the
body frame: x = mediolateral, y = anterior, z = superior. Bins are quintiles of the
GROUND-TRUTH knee flexion angle, so the binning variable is not itself estimated.

--- hip ---
   GT knee flexion       n    off x    off y    off z    |off|
       104-133 deg    2496     +0.0    -15.6     -8.2     36.4
       133-159 deg    2496     -0.1    -20.9    -10.1     38.6
       159-168 deg    2496     +0.4    -21.9    -14.7     38.7
       168-173 deg    2496     -1.4    -29.2    -13.9     42.9
       173-180 deg    2496     +0.7    -29.7    -11.3     43.0
   most-extended bin minus most-flexed,         x:   +0.69 mm   exact p = 0.7930   (n=11 participants)
   most-extended bin minus most-flexed,         y:  -13.60 mm   exact p = 0.0010   (n=11 participants)
   most-extended bin minus most-flexed,         z:   -1.95 mm   exact p = 0.1484   (n=11 participants)
   most-extended bin minus most-flexed,     |off|:   +6.19 mm   exact p = 0.0020   (n=11 participants)

--- knee ---
   GT knee flexion       n    off x    off y    off z    |off|   flex-sens
       104-133 deg    2496     +1.9     -7.7    +26.9     32.2       -16.6
       133-159 deg    2496     +1.6    -15.7    +22.8     32.3       -17.8
       159-168 deg    2496     +1.1    -18.2    +17.7     29.1       -17.3
       168-173 deg    2496     +0.6    -17.1    +16.5     28.1       -15.4
       173-180 deg    2496     +0.3    -18.0    +17.2     28.5       -10.1
   most-extended bin minus most-flexed,         x:   -1.51 mm   exact p = 0.0342   (n=11 participants)
   most-extended bin minus most-flexed,         y:  -11.67 mm   exact p = 0.0020   (n=11 participants)
   most-extended bin minus most-flexed,         z:   -8.97 mm   exact p = 0.0010   (n=11 participants)
   most-extended bin minus most-flexed,     |off|:   -2.82 mm   exact p = 0.0156   (n=11 participants)
   most-extended bin minus most-flexed,  flex-sens:   +6.17 mm   exact p = 0.0059   (n=11 participants)

--- ankle ---
   GT knee flexion       n    off x    off y    off z    |off|
       104-133 deg    2496     +2.3    -12.1     -0.0     19.1
       133-159 deg    2496     +1.4     -8.8     -6.1     16.3
       159-168 deg    2496     +0.8     -5.9     -6.2     13.6
       168-173 deg    2496     +0.1     -6.2     -5.8     14.0
       173-180 deg    2496     +0.6     -7.6     -3.6     14.6
   most-extended bin minus most-flexed,         x:   -1.48 mm   exact p = 0.0264   (n=11 participants)
   most-extended bin minus most-flexed,         y:   +3.97 mm   exact p = 0.0234   (n=11 participants)
   most-extended bin minus most-flexed,         z:   -3.71 mm   exact p = 0.0361   (n=11 participants)
   most-extended bin minus most-flexed,     |off|:   -4.27 mm   exact p = 0.0049   (n=11 participants)
```

S3d. Posture family with BH correction — `BIOCV_POSTURE_FDR.txt`

`|off|` is the offset magnitude, which is rotation-invariant, so binning it by flexion is not circular. The `flex-sens` row is the component resolved on the hip–ankle chord, whose orientation is set by the binning variable; that row is circular, is listed for completeness, and no claim in the manuscript rests on it. Quintiles run from most flexed (104–133°) to most extended (173–180°), and each row is the most-extended quintile minus the most-flexed.

```
BENJAMINI-HOCHBERG OVER THE POSTURE FAMILY
Family size m = 13: the highest-minus-lowest flexion-quintile difference for each offset
component at each lower-limb joint, exactly as run in BIOCV_POSTURE.txt.

|off| is the offset MAGNITUDE, which is rotation-invariant: binning it by flexion is not
circular. 'flex-sens' is the component resolved on the hip-ankle chord, whose orientation is
set by the flexion angle used to form the bins, so that row IS circular and is reported here
for completeness rather than as evidence.

  joint   component   diff mm        p     BH q  verdict
    hip           y    -13.60   0.0010   0.0065  SURVIVES
   knee           z     -8.97   0.0010   0.0065  SURVIVES
    hip       |off|     +6.19   0.0020   0.0065  SURVIVES
   knee           y    -11.67   0.0020   0.0065  SURVIVES
  ankle       |off|     -4.27   0.0049   0.0127  SURVIVES
   knee   flex-sens     +6.17   0.0059   0.0128  SURVIVES  [circular]
   knee       |off|     -2.82   0.0156   0.0290  SURVIVES
  ankle           y     +3.97   0.0234   0.0380  SURVIVES
  ankle           x     -1.48   0.0264   0.0381  SURVIVES
   knee           x     -1.51   0.0342   0.0427  SURVIVES
  ankle           z     -3.71   0.0361   0.0427  SURVIVES
    hip           z     -1.95   0.1484   0.1608  n.s.
    hip           x     +0.69   0.7930   0.7930  n.s.

11 of 13 survive BH; 11 were significant unadjusted.
```

S3e. Leave-one-out offset correction — `BIOCV_LOO.txt`

Two figures for the same gain circulate and are not the same quantity. The manuscript quotes +9.848 mm, the mean of eleven per-participant differences. The block below quotes +10.190 mm, the difference of frame-pooled means (28.493 against 18.303), which weights a participant by how many frames they contribute. Both were recomputed from the per-participant accumulator and agree with the values printed here; the relation is the same one that separates 5.062 from 5.086 in Table S8. Every declared verdict uses the per-participant form, because the participant is the inferential unit (Section 2.5).

```
T6 -- LEAVE-ONE-SUBJECT-OUT JOINT-CENTRE OFFSET CORRECTION, THEN RECOMPUTE ANGLES
11 subjects.
LOO = subject corrected using the mean offset of the OTHER 10 subjects (deployable).
in-sample = corrected with the subject's OWN offset (CIRCULAR; a ceiling, not a result).

=== A NESTED NULL FOR THE LEAVE-ONE-PARTICIPANT-OUT TABLE ===

Each contributing participant's mean offset vector is multiplied by a random +/-1 before
being averaged into the table, keeping its magnitude and destroying the shared direction.
500 draws. The statistic is the [pos:HKA] gain on the noRej_conf arm.

observed gain          +10.190 mm
nested null: mean      -0.598 mm, sd 6.580, range [-20.826, +15.176]
draws reaching the observed gain: 13 of 500

=== VERDICT, against the rule fixed above ===
The observed gain falls inside the nested null. The LOO rows must be reported without
p-values and Section 3.4 restated.

=== [pos:HKA] POSITION, THE QUANTITY THE DECLARED RULE JUDGES ===

         arm    raw mm  world tbl  body tbl  world gain  body gain
  noRej_conf    28.493     18.303    18.371     +10.190    +10.122
    rep_conf    28.471     18.803    18.884      +9.667     +9.587

=== VERDICT, against the rule fixed at the top of this file ===
World-frame table buys +10.190 mm, body-frame +10.122 mm, difference -0.068.
The two agree within a millimetre: the world-frame construction is not attenuating
the correction, and the published figure stands.

=== knee_flex ===
         arm      raw   LOO-corr  LOO body  in-sample         LOO gain (95% CI)    body gain
  noRej_conf    2.575      3.201     3.126      2.495   -0.626 [-0.874,-0.366]*       -0.550
    rep_conf    2.772      3.176     3.105      2.469   -0.404 [-0.665,-0.119]*       -0.333

=== knee_abd ===
         arm      raw   LOO-corr  LOO body  in-sample         LOO gain (95% CI)    body gain
  noRej_conf    2.637      2.768     2.826      2.486   -0.130 [-0.472,+0.236]        -0.188
    rep_conf    2.658      2.888     2.942      2.666   -0.230 [-0.516,+0.066]        -0.284

=== hip_flex ===
         arm      raw   LOO-corr  LOO body  in-sample         LOO gain (95% CI)    body gain
  noRej_conf    3.651      3.363     3.296      2.541   +0.288 [-0.512,+1.024]        +0.355
    rep_conf    3.845      3.387     3.342      2.562   +0.458 [-0.215,+1.080]        +0.503

THE IN-SAMPLE CEILING, TESTED. One mean difference per participant, exact sign-flip over 11.
Positive = correcting with the subject's OWN offset lowers the angle error. Still circular:
a ceiling on what any posture-blind per-participant translation fitted to POSITION could buy,
not an effect a study could deploy.

   outcome         arm  ceiling          exact 95% CI   exact p
 knee_flex  noRej_conf    0.166   [ -0.609,  +0.954]    0.6787
 knee_flex    rep_conf    0.382   [ -0.353,  +1.133]    0.2930
  knee_abd  noRej_conf    0.140   [ -0.238,  +0.558]    0.5195
```

knee_abd	rep_conf	-0.020	[-0.326, +0.316]	0.9043
hip_flex	noRej_conf	1.094	[+0.236, +2.029]	0.0098
hip_flex	rep_conf	1.260	[+0.416, +2.153]	0.0059

CEILING MINUS DEPLOYABLE TABLE, tested. Positive = the participant's OWN offset lowers the angle error by more than the population table does, i.e. what a distributed table leaves on the table. One mean difference per participant, exact sign-flip over 11.

outcome	arm	ceiling-LOO	exact 95% CI	exact p
knee_flex	noRej_conf	0.772	[+0.214, +1.368]	0.0098
knee_flex	rep_conf	0.767	[+0.206, +1.352]	0.0127
knee_abd	noRej_conf	0.289	[+0.104, +0.478]	0.0117
knee_abd	rep_conf	0.225	[+0.021, +0.433]	0.0312
hip_flex	noRej_conf	0.930	[+0.260, +1.661]	0.0078
hip_flex	rep_conf	0.906	[+0.294, +1.534]	0.0059

WHY A POPULATION TABLE HARMS THE ANGLE: does the harm track the angle change the own-versus-population offset mismatch predicts? Positive harm = the table RAISES the error. The predictor is the angle change obtained by applying the mismatch to the reference points, which is what a three-point angle actually responds to; a common translation cancels.

outcome	arm	n	mean pred	Pearson r	perm p
knee_flex	noRej_conf	11	2.187	-0.398	0.2267
knee_flex	rep_conf	11	2.220	-0.126	0.7095
hip_flex	noRej_conf	11	2.231	+0.308	0.3567
hip_flex	rep_conf	11	2.328	+0.283	0.3981

S3f. Offset geometry — `BIOCV_X22.txt`

The source for the axial / in-plane-perpendicular / out-of-plane decomposition in Table 3b, on which Section 3.4's account rests. The three components are means of per-frame magnitudes and therefore do not compose to the printed total.

```
X22 -- IS THE FIXED JOINT-CENTRE OFFSET COMPATIBLE WITH THE OBSERVED KNEE-ANGLE ERROR?
N=12480 knee-angle observations, 11 subjects.
bias-only angle = angle(GT + per-subject-per-joint systematic offset) - GT angle.

                          quantity   mean deg   median      p90
angle error from SYSTEMATIC offset      2.766    2.476    5.369
        angle error, full pipeline      2.722    2.243    5.698

RATIO OF MEAN ABSOLUTE ERRORS, bias-only / full pipeline: 101.6%
PER-FRAME CORRELATION between the two: r = 0.199

READ THIS PAIR TOGETHER, NOT THE FIRST LINE ALONE. The ratio says the systematic offset
is SUFFICIENT on its own to produce a knee-angle error of the observed size. It is not a
decomposition and it is not a share: both quantities are mean ABSOLUTE errors, which do
not add, so a ratio near 100% is compatible with the two errors being unrelated. The
correlation is what says whether they are the same error. Table 3a settles it from the
other side: correcting each subject with their OWN mean offset -- removing exactly the
quantity measured here -- lowers knee-flexion error from 2.575 to 2.495 deg, a recovery
of 0.080 deg, or 3% of the observed error. The static offset therefore BOUNDS the
magnitude of the angle error without accounting for it, and no causal claim may be built
on this table. What the offset does explain is why a triangulation gain of about a
millimetre need not reach the angle: the decomposition below shows only the in-plane
perpendicular component driving flexion at all.

--- why: per-frame decomposition of the knee offset relative to the hip-ankle chord ---
             component  mean |mm|     (flexion sensitivity)
    axial (along limb)      27.45  changes length, not flexion
in-plane perpendicular      11.04      DRIVES flexion error
          out-of-plane       8.63  ab/adduction, not flexion
    total |rel offset|      32.36

random (non-systematic) component of knee-angle error: -0.044 deg [-0.345,0.234] (subject-cluster bootstrap)
```

S3g. Not used.

S3h. Gauss–Newton contrast — `BIOCV_GN_DISSOCIATION.txt`

Rows with a counterpart in the $m = 50$ family carry that family's corrected q. Rows without one — the object-space rejection contrast under Gauss–Newton, and the hip-flexion and frontal-knee rows — belong to no declared family and are exploratory; the manuscript quotes none of them.

```
GAUSS-NEWTON REJECTION CONTRAST -- BASELINES AND RELATIVE EFFECTS

POSITIVE effect = the rejection arm has the LOWER error, matching the FAMILY tuple
(level, lm_noRej, lm_repRej) in biocv_permutation_v2.py. q-values are those of the m = 50
family in BIOCV_PERM_V3.txt; the p reprinted here is recomputed from the per-participant
means as a check that the same contrast is being described.

contrast                                                baseline    effect    rel %        p
[pos:HKA] GN: no rejection vs reprojection rejection      29.664    +0.858    +2.79   0.0020
[pos:HKA] GN: no rejection vs object-space rejection      29.664    +0.734    +2.40   0.0020
knee flexion, GN: no rejection vs reprojection             2.673    -0.169    -7.25   0.0146
knee flexion, GN: no rejection vs object-space             2.673    -0.124    -5.22   0.0049
hip flexion, GN: no rejection vs reprojection              4.047    -0.022    -0.93   0.7148
frontal knee, GN: no rejection vs reprojection             2.911    +0.052    +2.16   0.1895
```

S3i. Interaction family with BH correction — `BIOCV_INTERACTION_FDR.txt`

As in Table S3n, the block below is verbatim and still prints q-values on its leave-one-out offset-table rows. Those are withheld by the rule of Section 2.5; the manuscript's counts exclude them.

Table 2(c) of the main text lists all 20; they are repeated here with the unadjusted p-values alongside the corrected q-values. They are recomputed from the per-participant means dumped by the analysis scripts rather than transcribed from another table, so the correction runs on exact p-values.

```
BENJAMINI-HOCHBERG OVER THE INTERACTION FAMILY
Family size m = 20. Recomputed from the per-participant means in _persubj.pkl, _persubj_round5.pkl, _persubj_v2.pkl,
so the correction runs on exact sign-flip p-values rather than on values already rounded
to four decimals, matching the other three families.

Interaction = relative position effect minus relative angle effect, per participant, each
expressed as a fraction of that participant's own BASELINE error -- the arm named first in
the contrast. Position is measured at [pos:HKA], the hip, knee and ankle the angle is built
from.

intervention                                       dpos%   dang%   inter       p     BH q  verdict
criterion at matched k=1 [knee flex]               +0.07   -3.21   +3.29   0.0010   0.0049  SURVIVES
confidence->confidence x 1/d [hip flex]            -2.46   -6.58   +4.12   0.0010   0.0049  SURVIVES
LOO offset correction [knee flex]                 +34.04  -27.03  +61.07   0.0010   0.0049  SURVIVES
LOO offset correction [frontal knee]              +34.04  -10.31  +44.35   0.0010   0.0049  SURVIVES
confidence->confidence x 1/d [frontal knee]        -2.46   -7.67   +5.22   0.0020   0.0078  SURVIVES
LOO offset correction [hip flex]                  +34.04   -4.48  +38.52   0.0029   0.0098  SURVIVES
criterion at matched k=2 [knee flex]               +0.13   -4.01   +4.14   0.0039   0.0098  SURVIVES
GN: rejection (w^2-matched) [knee flex]            +3.45   -7.45  +10.90   0.0039   0.0098  SURVIVES
no rejection->objd rejection [knee flex]           +0.51   -6.09   +6.60   0.0059   0.0117  SURVIVES
GN: no rejection -> reproj rejection [knee flex]   +2.79   -7.25  +10.04   0.0059   0.0117  SURVIVES
no rejection->objd rejection [hip flex]            +0.51   -5.83   +6.34   0.0078   0.0142  SURVIVES
no rejection->reproj rejection [knee flex]         -0.10   -8.16   +8.07   0.0146   0.0244  SURVIVES
reproj->object-space criterion [frontal knee]      +0.61   -2.61   +3.22   0.0225   0.0346  SURVIVES
uniform->confidence weighting [hip flex]           +3.10   +5.69   -2.59   0.0498   0.0711  raw only
DLT -> GN w^2-matched [knee flex]                  -3.31   +0.22   -3.54   0.1611   0.2148  n.s.
criterion at matched k=3 [knee flex]               +0.58   -1.74   +2.31   0.1904   0.2380  n.s.
uniform->confidence weighting [knee flex]          +3.10   +1.43   +1.67   0.2412   0.2838  n.s.
reproj->object-space criterion [knee flex]         +0.61   +1.74   -1.13   0.3389   0.3765  n.s.
GN objective: w^2-matched -> unmatched (w) [knee flex]   -1.53   -1.03   -0.50   0.3887   0.4091  n.s.
uniform->confidence weighting [frontal knee]       +3.10   +3.06   +0.04   0.9473   0.9473  n.s.

13 of 20 survive BH; 14 were significant unadjusted.
Contrasts significant unadjusted that do not survive correction are exploratory.
```

S3j. Joint-set interaction — `BIOCV_JOINTSET_FDR.txt`

The test behind Table 1(c): whether an intervention moves the two joint sets by different fractions of their own baselines. It is computed from the same per-participant means as the m = 50 family rather than from an independent pass, which is one reason the declared families are not independent of one another (see S4).

```
DOES THE JOINT SET CHANGE THE EFFECT? -- [pos:AW] x [pos:HKA] INTERACTION

Whether an intervention moves the two joint sets by different amounts. Per participant, the effect on ankles
and wrists minus the effect on hip/knee/ankle, each as a percentage of that participant's
own baseline error at that joint set. Positive interaction = the intervention does more for
[pos:AW] than for [pos:HKA]. Exact sign-flip over 2^11 = 2048 patterns; BH within m = 8.

contrast                                  k     dAW%    dHKA%    inter        p     BH q  verdict
uniform vs confidence weighting          11   +23.41    +3.10   +20.31   0.0010   0.0020  SURVIVES
matched k=2: reprojection vs object-space  11    +2.54    +0.13    +2.41   0.0010   0.0020  SURVIVES
matched k=3: reprojection vs object-space  11    +4.49    +0.58    +3.91   0.0010   0.0020  SURVIVES
confidence vs confidence x 1/d weighting 11    +1.46    -2.46    +3.91   0.0010   0.0020  SURVIVES
adaptive: reprojection vs object-space   11    +1.95    +0.61    +1.34   0.0029   0.0047  SURVIVES
matched k=1: reprojection vs object-space  11    +0.92    +0.07    +0.84   0.0107   0.0143  SURVIVES
no rejection vs object-space rejection   11    +2.48    +0.51    +1.97   0.1504   0.1719  n.s.
no rejection vs reprojection rejection   11    +0.54    -0.10    +0.64   0.6182   0.6182  n.s.

6 of 8 survive BH; 6 were significant unadjusted.

HOW TO READ THIS. A surviving interaction means the joint set changes the size of the
effect and the dissociation is established directly. A null means we cannot demonstrate
that the two joint sets differ, and any claim that the joint set 'changes the answer' must
be stated as a change of verdict -- which is a fact about significance, hence about
precision as well as about effect size -- and not as a demonstrated difference in effect.
```

S3k. Exact 90% intervals on the angle contrasts — `BIOCV_EQUIV_BOUNDS.txt`

The exact 90% intervals quoted in Section 3.2, obtained by inverting the sign-flip test. Not every row is a null: three intervals here exclude zero, and Section 3.2 quotes only the bounds on the contrasts that are null under correction. The interaction family is reported in S3i.

```
EXACT EQUIVALENCE BOUNDS ON THE NULL CONTRASTS
11 subjects, 2^11 = 2048 sign patterns fully enumerated.
Intervals are EXACT sign-flip confidence intervals (test inversion), not bootstrap.

=== PART 1. EQUIVALENCE: how large an effect can we rule out? ===
90% CI on the per-subject mean difference. Under TOST, a contrast is equivalent to
zero at margin m if the whole 90% interval lies inside (-m, +m). We quote the
SMALLEST margin at which each null is declared equivalent -- the tightest honest
bound -- rather than testing one margin we chose ourselves.

contrast                                      effect               90% CI    bound  % of 5deg MDC
knee flex: criterion                          +0.047      [-0.007, +0.100]    0.100           2.0%
frontal knee: criterion                       -0.054      [-0.099, -0.008]    0.099           2.0%
hip flex: criterion                           +0.025      [-0.042, +0.090]    0.090           1.8%
knee flex: uniform vs confidence              +0.023      [-0.050, +0.095]    0.095           1.9%
frontal knee: uniform vs confidence           +0.078      [+0.056, +0.099]    0.099           2.0%
hip flex: uniform vs confidence               +0.227      [+0.153, +0.290]    0.290           5.8%
knee flex matched-k=3: criterion              -0.051      [-0.145, +0.045]    0.145           2.9%
knee flex: 1/d weighting                      +0.052      [-0.048, +0.159]    0.159           3.2%

The interaction family is reported in Table 2(c).
```

S3l. The upper-limb chain — `BIOCV_ANGLES.txt`

The shoulder–elbow–wrist chain, computed and excluded from the main analysis as non-gait-relevant. It is the one result that runs against the mechanism the Discussion proposes, so it is reproduced in full. These are subject-cluster bootstrap contrasts, not the per-participant exact sign-flip test used everywhere else in the paper.

```
=== ELBOW | n=12480 ===
 asym bin      n  noRej_conf    noRej_wd    rep_conf   objd_conf      rep_wd     objd_wd
    1.5-2   5918       12.01       12.15       11.76       11.90       12.00       12.09
    2-2.5   4570       12.07       12.11       11.84       11.90       11.90       12.04
    2.5-3   1626       12.41       12.45       11.68       11.91       12.19       12.58
      3-4    366       11.67       11.85       11.83       11.89       12.33       12.76
      ALL  12480       12.07       12.17       11.78       11.90       12.00       12.16
   does reproj rejection beat NO rejection?         rep_conf - noRej_conf = -0.295 deg [-0.494,-0.121]*
   does object-space rejection beat NO rejection?   objd_conf - noRej_conf = -0.172 deg [-0.385,+0.022]
   object-space vs reprojection criterion           rep_conf - objd_conf = -0.123 deg [-0.186,-0.066]*
   does 1/d triangulation weighting help?           rep_conf - rep_wd = -0.220 deg [-0.423,-0.027]*
   1/d weighting WITHOUT any rejection              noRej_conf - noRej_wd = -0.094 deg [-0.188,-0.002]*
   best-weighted: is rejection still harmful?       rep_wd - noRej_wd = -0.168 deg [-0.436,+0.086]
   no-rejection+1/d vs objd+1/d                     objd_wd - noRej_wd = -0.010 deg [-0.241,+0.218]
```

S3m. Bias–variance decomposition of joint-centre error — `BIOCV_GT_OFFSET.txt`

The per-participant offset means and standard deviations behind Section 3.4’s statement that between-participant variability is comparable to the mean.

```
B14 -- BIAS/VARIANCE DECOMPOSITION OF 3D JOINT ERROR
Arm = object-space rejection + confidence weights.
MSE = ||mean(e)||^2 (systematic offset) + mean(||e-mean(e)||^2) (random).

=== WORLD FRAME ===
       joint       n     RMSE     bias    resid  bias%MSE
  L-shoulder    6240     40.1     38.1     12.6       90%
  R-shoulder    6240     38.5     37.2      9.7       94%
     L-elbow    6240     28.4     24.6     14.3       75%
     R-elbow    6240     26.8     23.7     12.6       78%
     L-wrist    6240     20.4     16.5     12.0       65%
     R-wrist    6240     17.5     13.9     10.6       63%
       L-hip    6240     41.2     35.4     21.1       74%
       R-hip    6240     41.3     36.9     18.6       80%
      L-knee    6240     31.0     27.5     14.3       79%
      R-knee    6240     33.4     30.1     14.6       81%
     L-ankle    6240     17.4     11.5     13.0       44%
     R-ankle    6240     17.3     11.5     13.0       44%

THE SAME DECOMPOSITION WITH PARTICIPANTS POOLED, which is what a cohort-level validation
study measures. The common offset is the mean over participants of their mean error
vector; the between-participant scatter moves from the bias term into the random term.

       joint   common  scatter   random  within-ppt share  pooled share
  L-shoulder     36.6     11.0     12.6               90%           83%
  R-shoulder     34.6     14.3      9.7               94%           80%
     L-elbow     21.0     13.4     14.3               75%           54%
     R-elbow     21.6     10.1     12.6               78%           64%
     L-wrist     14.8      7.6     12.0               65%           52%
     R-wrist     12.2      7.0     10.6               63%           48%
       L-hip     28.5     21.9     21.1               74%           47%
       R-hip     31.6     20.0     18.6               80%           57%
      L-knee     24.4     13.4     14.3               79%           61%
      R-knee     28.2     11.2     14.6               81%           70%
     L-ankle      9.9      6.2     13.0               44%           32%
     R-ankle      9.5      6.7     13.0               44%           30%

         ALL   74880     30.9     27.4     14.2       79%

=== BODY FRAME ===
       joint       n     RMSE     bias    resid  bias%MSE
  L-shoulder    6240     40.1     37.9     13.0       89%
  R-shoulder    6240     38.5     37.1     10.3       93%
     L-elbow    6240     28.4     24.3     14.7       73%
     R-elbow    6240     26.8     23.5     13.0       76%
     L-wrist    6240     20.4     16.4     12.1       65%
     R-wrist    6240     17.5     13.8     10.7       63%
       L-hip    6240     41.2     35.4     20.9       74%
       R-hip    6240     41.3     36.9     18.5       80%
      L-knee    6240     31.0     27.5     14.4       78%
      R-knee    6240     33.4     30.0     14.9       80%
     L-ankle    6240     17.4     11.5     13.0       44%
     R-ankle    6240     17.3     11.5     13.0       44%

THE SAME DECOMPOSITION WITH PARTICIPANTS POOLED, which is what a cohort-level validation
study measures. The common offset is the mean over participants of their mean error
vector; the between-participant scatter moves from the bias term into the random term.

       joint   common  scatter   random  within-ppt share  pooled share
  L-shoulder     36.5     10.8     13.0               89%           82%
  R-shoulder     34.5     14.2     10.3               93%           79%
     L-elbow     20.7     13.4     14.7               73%           52%
```

```
     R-elbow     21.4     10.2     13.0               76%           63%
     L-wrist     14.8      7.5     12.1               65%           52%
     R-wrist     12.1      7.1     10.7               63%           47%
       L-hip     28.4     22.3     20.9               74%           46%
       R-hip     31.6     20.0     18.5               80%           57%
      L-knee     24.3     13.4     14.4               78%           60%
      R-knee     28.0     11.1     14.9               80%           70%
     L-ankle      9.9      6.2     13.0               44%           32%
     R-ankle      9.5      6.7     13.0               44%           30%

         ALL   74880     30.9     27.3     14.3       78%

=== IS THE OFFSET CONSISTENT ACROSS SUBJECTS? (body frame) ===
A definitional joint-centre offset should point the same way for every subject;
a per-subject calibration error should not.
       joint     mean offset (x=right,y=ant,z=up) mm  across-subj SD
  L-shoulder   [   +4.0,  -14.5,  -33.2]        [  6.9,  2.7,  7.3]
  R-shoulder   [  -14.1,  -15.2,  -27.6]        [  6.9,  8.9,  7.5]
     L-elbow   [   +2.0,  +16.5,  +12.3]        [  4.2,  6.8, 10.0]
     R-elbow   [   -0.1,  +16.4,  +13.7]        [  5.2,  5.0,  6.5]
     L-wrist   [   +7.9,  -11.6,   +4.7]        [  3.4,  2.7,  5.6]
     R-wrist   [   -4.7,  -10.5,   +3.6]        [  4.1,  3.8,  3.8]
       L-hip   [  -18.9,  -18.7,   -9.9]        [  7.9,  9.7, 17.2]
       R-hip   [  +18.5,  -22.9,  -11.4]        [  8.0,  7.0, 15.9]
      L-knee   [   -0.7,  -13.8,  +20.0]        [  2.8,  9.8,  7.8]
      R-knee   [   +2.9,  -16.1,  +22.7]        [  2.1,  6.5,  8.1]
     L-ankle   [   -1.7,   -8.8,   -4.2]        [  1.3,  3.3,  4.7]
     R-ankle   [   +3.6,   -8.1,   -3.5]        [  1.2,  4.2,  4.6]

=== THE MEDIOLATERAL SHARE OF THE SQUARED HIP OFFSET, FORMED PER PARTICIPANT ===

share = (mediolateral component)^2 / |offset|^2, computed for each participant and then
averaged. The ratio of the across-participant means is given beside it; the two differ
because the mediolateral component varies across participants by about 8 mm.

       joint   per-participant   ratio of means    n
       L-hip             33.3%            44.4%   11
       R-hip             27.9%            34.4%   11
      L-knee              2.9%             0.1%   11
      R-knee              1.3%             1.1%   11
```

S3n. Within-limb subset control — `BIOCV_SUBSET_CONTROL.txt`

The block below is reproduced verbatim from the artefact and therefore still prints q-values and SURVIVES verdicts on its leave-one-out offset-correction rows. Those verdicts are withdrawn by the rule of Section 2.5: each participant's correction is built from the other ten, so the eleven differences are not exchangeable and the sign-flip test does not apply. Table 1(d) marks the same rows ‡ withheld, and the counts quoted in the manuscript exclude them. The artefact is not edited, so that a reader can see exactly what was computed.

This analysis is reported here rather than in the Results because of what it can and cannot carry. A subset effect is the unweighted mean of its member joints, so the difference between two subsets is a fixed linear combination of the per-joint effects of Table 1(b): comparing {hip, knee, ankle} with {hip, ankle} asks whether the knee effect differs from the mean of the hip and ankle effects. The thirty pairwise differences therefore span ten free quantities, not thirty, and their survivor counts are not thirty findings. What the control does establish, with body region and occlusion regime held constant, is that the joint set a study summarises over changes the SIZE of the effect it reports. After both sensitivities that reduces to one deployable contrast, which is the one the Results quote; the verdict changes are on retention controls or are unlicensed by a tested difference.

The control behind Table 1(d). The paper's two joint sets differ in body region as well as membership, so this compares subsets within the lower limb, where region, occlusion regime and camera geometry are constant and only membership varies.

```
IS IT THE JOINT SET, OR IS IT ARMS VERSUS LEGS?

Subsets of the SAME lower limb, so body region, occlusion regime and camera geometry are held
constant and only the membership of the set changes. Every subset, including the full
three-joint one, is rebuilt from the same per-joint accumulators. Exact sign-flip over
2^11 = 2048 sign patterns throughout. 'dep' marks an intervention a validation study could
actually run: matched-k is a retention control and is not deployable.

=== PART A. DOES MEMBERSHIP CHANGE THE EFFECT? relative scale, BH within m = 30 ===
Each participant's effect as a percentage of that participant's own baseline for that subset;
the tested quantity is the difference between two subsets. Every pair is tested, so the
manuscript never has to compare two separate significance verdicts (Gelman and Stern, 2006).

intervention                  dep  subset A        subset B             A%      B%    diff        p     BH q  verdict
uniform vs confidence weighting  y  hip+knee+ankle  hip+ankle         +3.10   +4.15   -1.05   0.0010   0.0042  SURVIVES
criterion at matched k=1         n  hip+knee+ankle  hip+ankle         +0.05   +0.73   -0.67   0.0010   0.0042  SURVIVES
criterion at matched k=1         n  knee+ankle      hip+ankle         -0.45   +0.73   -1.18   0.0010   0.0042  SURVIVES
criterion at matched k=2         n  hip+knee+ankle  knee+ankle        +0.11   -0.97   +1.08   0.0010   0.0042  SURVIVES
criterion at matched k=2         n  hip+knee+ankle  hip+ankle         +0.11   +1.44   -1.33   0.0010   0.0042  SURVIVES
criterion at matched k=2         n  knee+ankle      hip+ankle         -0.97   +1.44   -2.41   0.0010   0.0042  SURVIVES
criterion at matched k=2         n  hip+knee        hip+ankle         -0.20   +1.44   -1.64   0.0010   0.0042  SURVIVES
uniform vs confidence weighting  y  hip+knee        hip+ankle         +2.68   +4.15   -1.47   0.0029   0.0098  SURVIVES
criterion at matched k=1         n  hip+knee        hip+ankle         -0.15   +0.73   -0.88   0.0029   0.0098  SURVIVES
uniform vs confidence weighting  y  knee+ankle      hip+ankle         +2.53   +4.15   -1.62   0.0098   0.0293  SURVIVES
LOO offset correction            y  hip+knee+ankle  hip+knee         +34.04  +37.25   -3.21   0.0117   0.0320  SURVIVES
criterion, adaptive rejection    y  hip+knee+ankle  knee+ankle        +0.61   +1.13   -0.53   0.0166   0.0383  SURVIVES
criterion, adaptive rejection    y  knee+ankle      hip+ankle         +1.13   +0.32   +0.81   0.0166   0.0383  SURVIVES
LOO offset correction            y  hip+knee        hip+ankle        +37.25  +28.49   +8.76   0.0186   0.0398  SURVIVES
criterion at matched k=1         n  hip+knee+ankle  knee+ankle        +0.05   -0.45   +0.51   0.0205   0.0403  SURVIVES
criterion, adaptive rejection    y  knee+ankle      hip+knee          +1.13   +0.45   +0.68   0.0215   0.0403  SURVIVES
criterion at matched k=2         n  knee+ankle      hip+knee          -0.97   -0.20   -0.78   0.0264   0.0465  SURVIVES
criterion, adaptive rejection    y  hip+knee+ankle  hip+ankle         +0.61   +0.32   +0.29   0.0371   0.0618  raw only
criterion at matched k=2         n  hip+knee+ankle  hip+knee          +0.11   -0.20   +0.31   0.0840   0.1297  n.s.
```

criterion, adaptive rejection	y	hip+knee+ankle	hip+knee	+0.61	+0.45	+0.15	0.0879	0.1297	n.s.
uniform vs confidence weighting	y	hip+knee+ankle	hip+knee	+3.10	+2.68	+0.42	0.0908	0.1297	n.s.
criterion at matched k=1	n	hip+knee+ankle	hip+knee	+0.05	-0.15	+0.20	0.1055	0.1438	n.s.
uniform vs confidence weighting	y	hip+knee+ankle	knee+ankle	+3.10	+2.53	+0.57	0.1533	0.2000	n.s.
LOO offset correction	y	hip+knee+ankle	hip+ankle	+34.04	+28.49	+5.55	0.2090	0.2612	n.s.
criterion at matched k=1	n	knee+ankle	hip+knee	-0.45	-0.15	-0.30	0.2959	0.3551	n.s.
criterion, adaptive rejection	y	hip+knee	hip+ankle	+0.45	+0.32	+0.13	0.3975	0.4579	n.s.
LOO offset correction	y	knee+ankle	hip+knee	+32.21	+37.25	-5.03	0.4121	0.4579	n.s.
LOO offset correction	y	knee+ankle	hip+ankle	+32.21	+28.49	+3.73	0.7129	0.7637	n.s.
LOO offset correction	y	hip+knee+ankle	knee+ankle	+34.04	+32.21	+1.82	0.7383	0.7637	n.s.
uniform vs confidence weighting	y	knee+ankle	hip+knee	+2.53	+2.68	-0.15	0.8027	0.8027	n.s.

17 of 30 survive BH.

=== PART B. THE SAME CONTRASTS IN mm, BH within m = 30 ===

Part A's null is equal FRACTIONAL change, which has no mechanical warrant: two subsets with different baselines can differ fractionally while the intervention moves them by identical absolute amounts. This part removes the denominator. A difference that survives in Part A but not here is a scale artefact rather than a membership effect.

intervention	dep	subset A	subset B	A mm	B mm	diff	p	BH q	verdict
uniform vs confidence weighting	y	hip+knee+ankle	hip+ankle	+0.917	+1.193	-0.276	0.0010	0.0042	SURVIVES
criterion at matched k=1	n	hip+knee+ankle	hip+ankle	+0.022	+0.193	-0.172	0.0010	0.0042	SURVIVES
criterion at matched k=1	n	knee+ankle	hip+ankle	-0.094	+0.193	-0.287	0.0010	0.0042	SURVIVES
criterion at matched k=2	n	hip+knee+ankle	knee+ankle	+0.031	-0.218	+0.249	0.0010	0.0042	SURVIVES
criterion at matched k=2	n	hip+knee+ankle	hip+ankle	+0.031	+0.374	-0.343	0.0010	0.0042	SURVIVES
criterion at matched k=2	n	knee+ankle	hip+ankle	-0.218	+0.374	-0.593	0.0010	0.0042	SURVIVES
criterion at matched k=2	n	hip+knee	hip+ankle	-0.063	+0.374	-0.437	0.0010	0.0042	SURVIVES
uniform vs confidence weighting	y	knee+ankle	hip+ankle	+0.594	+1.193	-0.599	0.0020	0.0059	SURVIVES
LOO offset correction	y	hip+knee+ankle	hip+knee	+9.848	+13.262	-3.414	0.0020	0.0059	SURVIVES
LOO offset correction	y	hip+knee	hip+ankle	+13.262	+8.419	+4.843	0.0020	0.0059	SURVIVES
criterion at matched k=1	n	hip+knee	hip+ankle	-0.035	+0.193	-0.228	0.0059	0.0160	SURVIVES
uniform vs confidence weighting	y	hip+knee+ankle	knee+ankle	+0.917	+0.594	+0.323	0.0078	0.0195	SURVIVES
LOO offset correction	y	knee+ankle	hip+knee	+7.862	+13.262	-5.400	0.0215	0.0496	SURVIVES
criterion at matched k=1	n	hip+knee+ankle	knee+ankle	+0.022	-0.094	+0.116	0.0283	0.0607	raw only
uniform vs confidence weighting	y	hip+knee	hip+ankle	+0.965	+1.193	-0.228	0.0332	0.0664	raw only
criterion, adaptive rejection	y	hip+knee+ankle	hip+ankle	+0.168	+0.083	+0.085	0.0381	0.0714	raw only
criterion, adaptive rejection	y	knee+ankle	hip+ankle	+0.264	+0.083	+0.182	0.0439	0.0776	raw only
uniform vs confidence weighting	y	knee+ankle	hip+knee	+0.594	+0.965	-0.371	0.0527	0.0879	n.s.
criterion, adaptive rejection	y	hip+knee+ankle	knee+ankle	+0.168	+0.264	-0.096	0.0625	0.0987	n.s.
criterion at matched k=2	n	knee+ankle	hip+knee	-0.218	-0.063	-0.156	0.1025	0.1493	n.s.
criterion, adaptive rejection	y	hip+knee	hip+ankle	+0.157	+0.083	+0.075	0.1045	0.1493	n.s.
criterion at matched k=2	n	hip+knee+ankle	hip+knee	+0.031	-0.063	+0.094	0.1250	0.1705	n.s.
criterion, adaptive rejection	y	knee+ankle	hip+knee	+0.264	+0.157	+0.107	0.1377	0.1796	n.s.
LOO offset correction	y	hip+knee+ankle	knee+ankle	+9.848	+7.862	+1.985	0.1895	0.2368	n.s.
LOO offset correction	y	hip+knee+ankle	hip+ankle	+9.848	+8.419	+1.429	0.2070	0.2484	n.s.
criterion at matched k=1	n	hip+knee+ankle	hip+knee	+0.022	-0.035	+0.056	0.2197	0.2535	n.s.
criterion at matched k=1	n	knee+ankle	hip+knee	-0.094	-0.035	-0.059	0.4658	0.5176	n.s.
uniform vs confidence weighting	y	hip+knee+ankle	hip+knee	+0.917	+0.965	-0.048	0.5996	0.6424	n.s.
criterion, adaptive rejection	y	hip+knee+ankle	hip+knee	+0.168	+0.157	+0.011	0.6748	0.6981	n.s.
LOO offset correction	y	knee+ankle	hip+ankle	+7.862	+8.419	-0.557	0.8223	0.8223	n.s.

13 of 30 survive BH.

Parts A and B agree on 22 of 30 contrasts.

Surviving on the relative scale only -- report these as scale-dependent, not as membership effects:

- uniform vs confidence weighting: hip+knee vs hip+ankle
- criterion, adaptive rejection: hip+knee+ankle vs knee+ankle
- criterion, adaptive rejection: knee+ankle vs hip+knee
- criterion, adaptive rejection: knee+ankle vs hip+ankle
- criterion at matched k=1: hip+knee+ankle vs knee+ankle
- criterion at matched k=2: knee+ankle vs hip+knee

=== PART C. EACH SUBSET'S OWN EFFECT, BH within m = 20 ===

What a validation study would report had it chosen that subset. Parts A and B test whether subsets DIFFER; this tests whether the intervention has an effect on each subset alone. A verdict that changes across rows of one intervention is the paper's claim in its most direct form -- and needs Part A to license it, since two verdicts are not a difference.

intervention	dep	subset	effect mm	rel %	p	BH q	verdict

```
uniform vs confidence weighting      y  hip+knee+ankle       +0.917   +3.10   0.0010   0.0024  SURVIVES
uniform vs confidence weighting      y  knee+ankle           +0.594   +2.53   0.0010   0.0024  SURVIVES
uniform vs confidence weighting      y  hip+knee             +0.965   +2.68   0.0010   0.0024  SURVIVES
uniform vs confidence weighting      y  hip+ankle            +1.193   +4.15   0.0010   0.0024  SURVIVES
criterion, adaptive rejection        y  hip+knee+ankle       +0.168   +0.61   0.0078   0.0142  SURVIVES
criterion, adaptive rejection        y  knee+ankle           +0.264   +1.13   0.0010   0.0024  SURVIVES
criterion, adaptive rejection        y  hip+knee             +0.157   +0.45   0.0098   0.0150  SURVIVES
criterion, adaptive rejection        y  hip+ankle            +0.083   +0.32   0.2695   0.3369  n.s.
criterion at matched k=1             n  hip+knee+ankle       +0.022   +0.05   0.7061   0.7061  n.s.
criterion at matched k=1             n  knee+ankle           -0.094   -0.45   0.0459   0.0612  raw only
criterion at matched k=1             n  hip+knee             -0.035   -0.15   0.6797   0.7061  n.s.
criterion at matched k=1             n  hip+ankle            +0.193   +0.73   0.0332   0.0474  SURVIVES
criterion at matched k=2             n  hip+knee+ankle       +0.031   +0.11   0.6543   0.7061  n.s.
criterion at matched k=2             n  knee+ankle           -0.218   -0.97   0.0098   0.0150  SURVIVES
criterion at matched k=2             n  hip+knee             -0.063   -0.20   0.5352   0.6296  n.s.
criterion at matched k=2             n  hip+ankle            +0.374   +1.44   0.0010   0.0024  SURVIVES
LOO offset correction                y  hip+knee+ankle       +9.848  +34.04   0.0010   0.0024  SURVIVES
LOO offset correction                y  knee+ankle           +7.862  +32.21   0.0020   0.0043  SURVIVES
LOO offset correction                y  hip+knee            +13.262  +37.25   0.0010   0.0024  SURVIVES
LOO offset correction                y  hip+ankle            +8.419  +28.49   0.0029   0.0059  SURVIVES

14 of 20 survive BH.

VERDICT CHANGES WITHIN THE LOWER LIMB (anatomy constant, only membership varies):
  [deployable] criterion, adaptive rejection
      significant on: hip+knee+ankle, knee+ankle, hip+knee
      null on:        hip+ankle
  [control only] criterion at matched k=1
      significant on: hip+ankle
      null on:        hip+knee+ankle, knee+ankle, hip+knee
  [control only] criterion at matched k=2
      significant on: knee+ankle, hip+ankle
      null on:        hip+knee+ankle, hip+knee

HOW TO READ THIS. A surviving difference in Part A AND Part B means that dropping one joint
from a lower-limb set changes the reported effect with anatomy held constant -- membership is
doing the work, not upper-versus-lower limb. A verdict change in Part C on a DEPLOYABLE
intervention bears on practice; a verdict change on matched-k arms, which are controls,
shows the artefact rather than the behaviour of a deployed rule.
```

Table S4. Pooled-family sensitivity

Splitting a set of tests into separately corrected families can inflate survival. This is the consequence of the strictest alternative: one Benjamini–Hochberg correction over the union of all 14 declared families (m = 244). The counts are in the block below. The families are not independent — they are analyses of the same 6,240 frames, and the joint-set and interaction families are re-parameterisations of rows already in the m = 50 family — so the pooled correction double-counts and the split is a presentational choice rather than a looser error-rate convention. Against the declared-family baseline of 144 survivors the pooled count falls to 141. The declared families hold 247 contrasts; every count here runs over the 244 of them that carry a verdict. Three do not: two are held by the endpoint veto declared with the shank-foot outcome, and one is the leave-one-out row of that same family, whose q Section 2.5 withholds. Five verdicts change under pooling, one gaining and four losing, and none is a headline claim. The block below also reconciles two counts that are not the same number — the verdict each artefact printed, and a Benjamini-Hochberg recomputation from the raw p within the family declared here — and names the contrasts where they differ.

```
POOLING ALL 14 INFERENTIAL FAMILIES INTO ONE BH CORRECTION

The manuscript corrects within 14 declared families. This is the consequence of the
alternative convention -- one correction over their union -- reported so that a reader can
judge whether the split inflates survival.

  main contrasts               m = 50
  position vs angle            m = 20
  per-joint                    m = 12
  posture                      m = 13
  joint set vs joint set       m = 8
  within-limb: membership, %   m = 30
  within-limb: membership, mm  m = 30
  dose-response                m = 6
  within-limb: subset alone    m = 20
  offset control: per-joint    m = 18
  offset control: per-subset   m = 16
  offset control: reversal test m = 18
  estimator x rejection        m = 2
  shank-foot angle             m = 1
  POOLED                       m = 244

5 of 244 contrasts change verdict when pooled.

family                        contrast                                         p    own q  pooled q
within-limb: membership, mm   criterion at matched k=1                    0.0283   0.0607   0.0490  gains
within-limb: subset alone     criterion at matched k=1                    0.0332   0.0474   0.0563  loses
offset control: reversal test adaptive criterion knee                     0.0332   0.0460   0.0563  loses
posture                       x                                           0.0342   0.0427   0.0576  loses
posture                       z                                           0.0361   0.0427   0.0595  loses

A row whose q RISES above 0.05 under pooling loses a verdict its own family granted,
A row whose q FALLS below 0.05 gains a verdict the split correction withheld. Compare
each row's own q with its pooled q: the two directions are not interchangeable.

survivors, as each artefact printed the verdict:           144
survivors, recomputing BH from p within the declared family: 144
survivors under one pooled correction over m = 244:           141


2 contrast(s) carry NO VERDICT in their artefact because a rule declared with
the outcome withheld one, not because any correction disagreed. They are excluded from
every count above:
  shank-foot angle            none vs adaptive reprojection rejection   p=0.0010  own q=0.0020  artefact: NO VERDICT: heel disagrees in sign
```

```
  shank-foot angle            none vs adaptive object-space rejection   p=0.0010  own q=0.0020  artefact: NO VERDICT: heel disagrees in sign
```

CAVEAT ON POOLING. The declared families test different questions on partly overlapping observations: the per-joint family excludes the pooled [pos:HKA] rows precisely because those belong to the main family, and the joint-set interactions are computed from the same per-participant means as the main family rather than from an independent pass. Pooling therefore double-counts information and is conservative in a way that is not principled, which is why the manuscript does not adopt it. It is reported because a reader is entitled to see what the stricter convention would do.

Table S4b. Declared families

Section 2.5 corrects within each of these, never across them. The nine that predate the offset control total m = 189; the union the pooled sensitivity of Table S4 corrects over is all fourteen, m = 244, including the two families declared with the experiments they belong to (Tables S11 and S13). Both sensitivities cover the two offset-control families of Table S7, pooling their two reference arms, which is stricter than the manuscript's own correction.

family	m	where
main contrasts	50	Table S1
position against angle	20	Table S3i
joint set against joint set	8	Table S3j
per-joint	12	Table S3b
posture	13	Table S3d
dose-response	6	Table 2b
within-limb membership, relative	30	Table S3n Part A
within-limb membership, mm	30	Table S3n Part B
within-limb per-subset effects	20	Table S3n Part C
offset control, per-joint	6 contrasts x 3 joints; BH applied within m = 18 per reference column, verdicts taken from the midpoint	Table S7 Part A
offset control, per-subset	4 x 2 retentions x 2 arms; BH applied within m = 8 per arm	Table S7 Part C
estimator x rejection	2	Table S11
shank-foot angle	4	Table S13
offset control, reversal test	6 contrasts x 3 joints; the difference between the as-measured and offset-removed effect, BH within m = 18	Table S7 Part G

The m values above are the families as first declared. Every family containing leave-one-out rows was corrected again without them, because a q ranked against a p-value that does not apply is not a valid q. The reduced families are 46, 17, 24, 24 and 16 in place of 50, 20, 30, 30 and 20 (Table S22); the shank-foot family holds a leave-one-out row too, but Table S22 does not recompute it, because the endpoint veto leaves only one row carrying a verdict either way.

The family sizes printed in Table S4 count only the contrasts that carry a verdict, so the shank-foot family appears there as m = 1, against the 3 of Table S5, which drops only the leave-one-out row whose q is withheld, and the 4 declared with the outcome in Table S13. The two rows held by the endpoint veto are the difference between 3 and 1.

Four counts follow from these families and are not interchangeable. Benjamini-Hochberg within each family as first declared returns 146 verdicts (Table S5). Two shank-foot rows carry no verdict because an endpoint veto declared with the outcome withheld one, leaving 144 (Table S4). Re-correcting on the reduced families moves one verdict, giving 143, which is the count the manuscript reports (Table S22). A single pooled correction over the union m = 244, an alternative to declaring families at all, gives 141 (Table S4). The arbitrary-dependence sensitivity leaves 107 of the 146 (Table S5).

A note on sign, for cross-checking. Every table names its two arms in the order they are subtracted, so a contrast printed here as "A vs B" appears in the main tables as "B vs A" with the opposite sign wherever the main table names them the other way round. Both are correct under their own stated convention; compare the arm ORDER before concluding that a sign disagrees.

Table S5. Arbitrary-dependence sensitivity

Section 2.5 corrects with Benjamini–Hochberg, whose proof assumes independent test statistics, and appeals to the extension to positive regression dependence. That assumption is not verifiable for this design, so the price of dropping it is reported here: the same authors' procedure for arbitrary dependence divides the threshold by the harmonic number H_m. It is deliberately conservative and is not the convention adopted, but it shows exactly which verdicts rest on the positive-dependence assumption.

```
BENJAMINI-YEKUTIELI SENSITIVITY: THE PRICE OF NOT ASSUMING POSITIVE DEPENDENCE

Section 2.5 corrects with Benjamini-Hochberg within each declared family and appeals to the
positive-regression-dependence extension (Benjamini and Yekutieli, 2001) because the
contrasts in a family share arms, participants and baselines. PRDS is assumed there, not
shown. This is what the same authors' ARBITRARY-dependence procedure gives instead: the BH
threshold divided by H_m = sum_{i=1..m} 1/i. It is deliberately conservative and is NOT the
convention the manuscript adopts; it is reported so a reader can see which verdicts depend on
the positive-dependence assumption.

family                          m    H_m    BH    BY  contrasts losing significance under BY
main contrasts                 50   4.50    27    18  9
position vs angle              20   3.60    13    10  3
joint set vs joint set          8   2.72     6     6  0
per-joint                      12   3.10     7     7  0
posture                        13   3.18    11     6  5
dose-response                   6   2.45     6     6  0
estimator x rejection           2   1.50     1     1  0
shank-foot angle                3   1.83     2     2  0
within-limb: membership, %     30   3.99    17     9  8
within-limb: membership, mm    30   3.99    13    10  3
within-limb: subset alone      20   3.60    14    10  4
offset control: per-joint      18   3.50     8     7  1
offset control: per-subset     16   3.38     8     7  1
offset control: reversal test  18   3.50    13     8  5

TOTAL                                      146   107

Verdicts that hold under BH and not under BY, i.e. those resting on the positive-dependence
assumption:
  main contrasts               [pos:AW] does 1/d refine confidence? 11
  main contrasts               [pos:HKA] reprojection vs object-space criterion 11
  main contrasts               knee flex: reproj rejection vs none 11
  main contrasts               hip flex: reproj rejection vs none 11
  main contrasts               hip flex: objd rejection vs none 11
  main contrasts               knee flex matched-k=1: reprojection vs object-space at equal retention 11
  main contrasts               knee flex matched-k=2: reprojection vs object-space at equal retention 11
  main contrasts               knee flex: LOO population offset correction 11
  main contrasts               knee flex GN: no rejection vs reprojection rejection 11
  position vs angle            no rejection->objd rejection [hip flex] +0.51 -5.83
  position vs angle            no rejection->reproj rejection [knee flex] -0.10 -8.16
  position vs angle            reproj->object-space criterion [frontal knee] +0.61 -2.61
  posture                      knee |off|
  posture                      ankle y
  posture                      ankle x
  posture                      knee x
  posture                      ankle z
  within-limb: membership, %   uniform vs confidence weighting y knee+ankle hip+ankle +2.53 +4.15
  within-limb: membership, %   LOO offset correction y hip+knee+ankle hip+knee +34.04 +37.25
  within-limb: membership, %   criterion, adaptive rejection y hip+knee+ankle knee+ankle +0.61 +1.13
  within-limb: membership, %   criterion, adaptive rejection y knee+ankle hip+ankle +1.13 +0.32
  within-limb: membership, %   LOO offset correction y hip+knee hip+ankle +37.25 +28.49
  within-limb: membership, %   criterion at matched k=1 n hip+knee+ankle knee+ankle +0.05 -0.45
  within-limb: membership, %   criterion, adaptive rejection y knee+ankle hip+knee +1.13 +0.45
  within-limb: membership, %   criterion at matched k=2 n knee+ankle hip+knee -0.97 -0.20
  within-limb: membership, mm  criterion at matched k=1 n hip+knee hip+ankle -0.035 +0.193
```

```
  within-limb: membership, mm   uniform vs confidence weighting y hip+knee+ankle knee+ankle +0.917 +0.594
  within-limb: membership, mm   LOO offset correction y knee+ankle hip+knee +7.862 +13.262
  within-limb: subset alone     criterion, adaptive rejection y hip+knee+ankle +0.168
  within-limb: subset alone     criterion, adaptive rejection y hip+knee +0.157
  within-limb: subset alone     criterion at matched k=1 n hip+ankle +0.193
  within-limb: subset alone     criterion at matched k=2 n knee+ankle -0.218
  offset control: per-joint     uniform -> confidence hip midpoint
  offset control: per-subset    hip+knee -0.063 2nd arm
  offset control: reversal test adaptive criterion hip -0.024 +0.264
  offset control: reversal test adaptive criterion knee +0.339 +0.181
  offset control: reversal test uniform -> confidence ankle +0.822 +0.912
  offset control: reversal test no rejection -> object-space rejection knee -0.546 -0.020
  offset control: reversal test no rejection -> reprojection rejection knee -0.885 -0.203

Verdicts holding under BOTH are those whose p reaches the sign-flip floor or close to it, and
are unaffected by which dependence assumption is made.
```

Table S6. Leave-one-participant-out robustness

With eleven participants and a sign-flip floor of 0.000977, a contrast at that floor means all eleven agreed in direction, and the obvious question is whether one of them carries the result. Each contrast the paper rests on was refitted eleven times, dropping each participant in turn and re-running the exact test on the remaining ten.

LEAVE-ONE-PARTICIPANT-OUT ROBUSTNESS OF THE HEADLINE CONTRASTS

Each participant dropped in turn; the exact sign-flip test re-run on the remaining ten (2^10 = 1024 patterns, smallest attainable p = 0.001953). 'worst p' is the largest p over the eleven refits and 'driver' the participant whose removal produces it. A contrast is robust when the direction is unchanged in all eleven refits and the worst p stays below 0.05.

contrast	effect	p	LOO min	LOO max	worst p	driver	verdict
[pos:HKA] uniform vs confidence weighting	+0.917	0.0010	+0.832	+0.967	0.0020	P03	robust
[pos:HKA] criterion, adaptive rejection	+0.168	0.0078	+0.148	+0.190	0.0156	P04	robust
[pos:HKA] LOO offset correction	+9.848	0.0010	+9.328	+10.643	0.0020	P03	robust
[pos:HKA] GN: no rejection vs reproj rejection	+0.858	0.0020	+0.702	+0.958	0.0039	P03	robust
hip, criterion at matched k=2	+0.530	0.0020	+0.432	+0.586	0.0039	P04	robust
knee, criterion at matched k=2	-0.655	0.0020	-0.734	-0.575	0.0039	P03	robust
knee, criterion, adaptive rejection	+0.339	0.0020	+0.312	+0.378	0.0039	P03	robust
hip, uniform vs confidence weighting	+1.564	0.0010	+1.437	+1.678	0.0020	P03	robust
knee flexion, LOO offset correction	-0.606	0.0068	-0.676	-0.540	0.0137	P03	robust
knee flexion, GN: no rejection vs reproj	-0.169	0.0146	-0.198	-0.143	0.0293	P03	robust
knee flexion, adaptive reprojection rejection	-0.188	0.0098	-0.221	-0.164	0.0195	P03	robust
frontal knee, uniform vs confidence weighting	+0.078	0.0010	+0.070	+0.083	0.0020	P03	robust
hip flexion, uniform vs confidence weighting	+0.227	0.0020	+0.213	+0.265	0.0039	P03	robust
{knee,ankle} vs {hip,ankle} at matched k=2 (mm)	-0.593	0.0010	-0.634	-0.550	0.0020	P03	robust

14 of 14 headline contrasts are robust to dropping any single participant.

No headline result depends on any one participant.

Table S7. Is the per-joint sign structure an artefact of offset direction?

Table S3m shows that 74–81% of the position error at hip and knee is a fixed joint-centre offset of 27–37 mm, against per-joint effects of 0.2–0.7 mm. A scalar error measured against a displaced target changes only through the perturbation's component along the offset, so a per-joint sign could report offset direction rather than triangulation accuracy. The twelve per-joint contrasts of Table 1b were therefore repeated on offset-removed error, subtracting each participant's own mean error vector for that joint from both arms of a contrast. Which arm supplies the offset is a free choice, so both are reported. Part B splits the same effects into components along and perpendicular to the offset.

What the parts contain. (A) the twelve per-joint contrasts on offset-removed error, plus the two rejection contrasts; (B) the split into components along and perpendicular to the offset; (D) the offset estimated out of sample; (E) superseded, see (F); (C) the matched-k subset control; (F) dispersion against centroid. They print in the order they are computed, which is not alphabetical. (G) tests the reversal the title asserts as a DIFFERENCE between the as-measured and offset-removed effects, rather than as a pair of verdicts, which is the standard Section 2.5 applies to every other such claim; its reading was fixed before the numbers existed.

Parts E and F, in order. Part E asked whether the offset-removed effect is a residual mean shift or a change in dispersion, and it cannot answer: it subtracts the MIDPOINT of the two arms' mean errors from both, which leaves residual means of equal norm and opposite sign, so its mean-shift column is zero for every row by construction. It is printed because it was run and reported, and it is withdrawn. Part F is the test that can fail, on quantities that are both free to differ, and it is the one the manuscript cites.

```
IS THE PER-JOINT SIGN STRUCTURE AN ARTEFACT OF OFFSET DIRECTION?

11 subjects -> 2^11 = 2048 sign patterns, fully enumerated.
POSITIVE effect = the SECOND-NAMED arm is better (effect = mean|e| of the first arm minus
mean|e| of the second), the same convention as every other family in this paper.
Error vectors are expressed in a body frame (x right, y anterior, z up).

=== (A) THE SAME CONTRASTS ON OFFSET-REMOVED ERROR ===

|e - b| where b is that participant's own mean error vector for the joint, taken from ONE
reference arm and subtracted from BOTH arms. 'as measured' repeats the published per-joint
effect for comparison. If the hip/knee signs oppose in the offset-removed columns too, the
dissociation is a property of triangulation and not of where each joint's offset points.

Verdicts come from the MIDPOINT column. The two single-arm columns are the endpoints of a
known downward/upward bias (see the note above pair_scalars) and are shown for reference
only; a range must not be built from them.

contrast                    joint  as measured    p meas     midpoint         p  ref=1st arm  ref=2nd arm
adaptive criterion            hip       -0.024    0.8350       +0.264    0.0010       +0.221       +0.308
adaptive criterion           knee       +0.339    0.0020       +0.181    0.0547       +0.170       +0.192
adaptive criterion          ankle       +0.190    0.0352       +0.150    0.1084       +0.142       +0.158

criterion at matched k=1      hip       +0.253    0.0840       -0.718    0.0010       -0.800       -0.632
criterion at matched k=1     knee       -0.322    0.0029       +0.399    0.0039       +0.372       +0.426
criterion at matched k=1    ankle       +0.134    0.1484       +0.076    0.3066       +0.063       +0.088

criterion at matched k=2      hip       +0.530    0.0020       -0.108    0.5771       -0.222       +0.009
criterion at matched k=2     knee       -0.655    0.0020       +0.738    0.0029       +0.643       +0.828
criterion at matched k=2    ankle       +0.219    0.0762       +0.215    0.0664       +0.181       +0.247
```

uniform -> confidence	hip	+1.564	0.0010	+0.253	0.0088	+0.069	+0.435
uniform -> confidence	knee	+0.365	0.0010	-0.024	0.7266	-0.031	-0.017
uniform -> confidence	ankle	+0.822	0.0010	+0.912	0.0010	+0.903	+0.921
no rejection -> object-space rejection	hip	+0.608	0.0234	-1.726	0.0010	-1.878	-1.564
no rejection -> object-space rejection	knee	-0.546	0.0039	-0.020	0.8486	-0.086	+0.046
no rejection -> object-space rejection	ankle	+0.426	0.0039	+0.256	0.1621	+0.213	+0.299
no rejection -> reprojection rejection	hip	+0.633	0.0439	-1.983	0.0010	-2.240	-1.714
no rejection -> reprojection rejection	knee	-0.885	0.0010	-0.203	0.2051	-0.279	-0.125
no rejection -> reprojection rejection	ankle	+0.237	0.1738	+0.107	0.6035	+0.084	+0.130

Benjamini-Hochberg within m = 18, run twice -- once per reference arm -- so each column is a complete correction over the same contrasts, not one correction over both. The first four contrasts are Table 1b's; the last two are the rejection contrasts the Conclusion's position null rests on, added under the rule declared at the top of this file.

contrast	joint	reference	effect mm	p	q	verdict
adaptive criterion	hip	midpoint	+0.264	0.0010	0.0035	SURVIVES
adaptive criterion	hip	1st arm	+0.221	0.0322	0.0829	n.s.
adaptive criterion	hip	2nd arm	+0.308	0.0010	0.0035	SURVIVES
adaptive criterion	knee	midpoint	+0.181	0.0547	0.1094	n.s.
adaptive criterion	knee	1st arm	+0.170	0.0645	0.1450	n.s.
adaptive criterion	knee	2nd arm	+0.192	0.0449	0.0809	n.s.
adaptive criterion	ankle	midpoint	+0.150	0.1084	0.1774	n.s.
adaptive criterion	ankle	1st arm	+0.142	0.1221	0.1998	n.s.
adaptive criterion	ankle	2nd arm	+0.158	0.0879	0.1438	n.s.
criterion at matched k=1	hip	midpoint	-0.718	0.0010	0.0035	SURVIVES
criterion at matched k=1	hip	1st arm	-0.800	0.0010	0.0044	SURVIVES
criterion at matched k=1	hip	2nd arm	-0.632	0.0010	0.0035	SURVIVES
criterion at matched k=1	knee	midpoint	+0.399	0.0039	0.0100	SURVIVES
criterion at matched k=1	knee	1st arm	+0.372	0.0049	0.0146	SURVIVES
criterion at matched k=1	knee	2nd arm	+0.426	0.0039	0.0088	SURVIVES
criterion at matched k=1	ankle	midpoint	+0.076	0.3066	0.3943	n.s.
criterion at matched k=1	ankle	1st arm	+0.063	0.3828	0.4594	n.s.
criterion at matched k=1	ankle	2nd arm	+0.088	0.2295	0.3178	n.s.
criterion at matched k=2	hip	midpoint	-0.108	0.5771	0.6790	n.s.
criterion at matched k=2	hip	1st arm	-0.222	0.2930	0.4056	n.s.
criterion at matched k=2	hip	2nd arm	+0.009	0.9619	0.9619	n.s.
criterion at matched k=2	knee	midpoint	+0.738	0.0029	0.0088	SURVIVES
criterion at matched k=2	knee	1st arm	+0.643	0.0049	0.0146	SURVIVES
criterion at matched k=2	knee	2nd arm	+0.828	0.0020	0.0050	SURVIVES
criterion at matched k=2	ankle	midpoint	+0.215	0.0664	0.1195	n.s.
criterion at matched k=2	ankle	1st arm	+0.181	0.1191	0.1998	n.s.
criterion at matched k=2	ankle	2nd arm	+0.247	0.0391	0.0781	n.s.
uniform -> confidence	hip	midpoint	+0.253	0.0088	0.0198	SURVIVES
uniform -> confidence	hip	1st arm	+0.069	0.3340	0.4294	n.s.
uniform -> confidence	hip	2nd arm	+0.435	0.0020	0.0050	SURVIVES
uniform -> confidence	knee	midpoint	-0.024	0.7266	0.7693	n.s.
uniform -> confidence	knee	1st arm	-0.031	0.6543	0.6787	n.s.
uniform -> confidence	knee	2nd arm	-0.017	0.8018	0.8489	n.s.
uniform -> confidence	ankle	midpoint	+0.912	0.0010	0.0035	SURVIVES
uniform -> confidence	ankle	1st arm	+0.903	0.0010	0.0044	SURVIVES
uniform -> confidence	ankle	2nd arm	+0.921	0.0010	0.0035	SURVIVES
no rejection -> object-space rejection	hip	midpoint	-1.726	0.0010	0.0035	SURVIVES
no rejection -> object-space rejection	hip	1st arm	-1.878	0.0010	0.0044	SURVIVES
no rejection -> object-space rejection	hip	2nd arm	-1.564	0.0010	0.0035	SURVIVES
no rejection -> object-space rejection	knee	midpoint	-0.020	0.8486	0.8486	n.s.
no rejection -> object-space rejection	knee	1st arm	-0.086	0.4336	0.4878	n.s.
no rejection -> object-space rejection	knee	2nd arm	+0.046	0.6504	0.7317	n.s.
no rejection -> object-space rejection	ankle	midpoint	+0.256	0.1621	0.2432	n.s.
no rejection -> object-space rejection	ankle	1st arm	+0.213	0.2539	0.3809	n.s.
no rejection -> object-space rejection	ankle	2nd arm	+0.299	0.1123	0.1685	n.s.
no rejection -> reprojection rejection	hip	midpoint	-1.983	0.0010	0.0035	SURVIVES
no rejection -> reprojection rejection	hip	1st arm	-2.240	0.0010	0.0044	SURVIVES
no rejection -> reprojection rejection	hip	2nd arm	-1.714	0.0010	0.0035	SURVIVES
no rejection -> reprojection rejection	knee	midpoint	-0.203	0.2051	0.2840	n.s.
no rejection -> reprojection rejection	knee	1st arm	-0.279	0.0986	0.1973	n.s.
no rejection -> reprojection rejection	knee	2nd arm	-0.125	0.4219	0.5424	n.s.
no rejection -> reprojection rejection	ankle	midpoint	+0.107	0.6035	0.6790	n.s.

no rejection -> reprojection rejection	ankle	1st arm	+0.084	0.6787	0.6787	n.s.
no rejection -> reprojection rejection	ankle	2nd arm	+0.130	0.5205	0.6246	n.s.

No row has a midpoint verdict differing from both single-arm verdicts.

=== (B) HOW MUCH OF EACH EFFECT IS MOTION ALONG THE OFFSET DIRECTION? ===

For each participant, dvec = mean(e_first) - mean(e_second) and bhat = the unit mean error vector of the first-named arm. To first order the scalar effect equals bhat . dvec, so the 'along' column is the linearised prediction of the 'as measured' column. 'share' is along / as-measured: near 1 means the reported effect is entirely offset-mediated.

contrast	joint	|b| mm	as measured	along	perp	share
adaptive criterion	hip	34.67	-0.024	-0.286	1.051	11.80
adaptive criterion	knee	28.27	+0.339	+0.295	0.365	0.87
adaptive criterion	ankle	11.25	+0.190	+0.162	0.420	0.86
criterion at matched k=1	hip	33.82	+0.253	+0.594	1.329	2.35
criterion at matched k=1	knee	28.27	-0.322	-0.488	0.651	1.52
criterion at matched k=1	ankle	11.19	+0.134	+0.202	0.445	1.51
criterion at matched k=2	hip	32.78	+0.530	+0.810	1.688	1.53
criterion at matched k=2	knee	28.57	-0.655	-0.987	1.355	1.51
criterion at matched k=2	ankle	11.33	+0.219	+0.276	0.832	1.26
uniform -> confidence	hip	38.30	+1.564	+1.726	1.692	1.10
uniform -> confidence	knee	27.70	+0.365	+0.289	0.235	0.79
uniform -> confidence	ankle	10.57	+0.822	-0.444	0.621	-0.54
no rejection -> object-space rejection	hip	36.63	+0.608	+1.697	1.495	2.79
no rejection -> object-space rejection	knee	27.41	-0.546	-0.540	1.038	0.99
no rejection -> object-space rejection	ankle	11.07	+0.426	-0.007	0.774	-0.02
no rejection -> reprojection rejection	hip	36.63	+0.633	+2.039	1.782	3.22
no rejection -> reprojection rejection	knee	27.41	-0.885	-0.829	1.032	0.94
no rejection -> reprojection rejection	ankle	11.07	+0.237	-0.164	0.640	-0.69

=== (D) THE SAME CONTRASTS WITH THE OFFSET ESTIMATED OUT OF SAMPLE ===

Three removals, weakest first. ODD/EVEN splits retained frames within a trial, which are 90 ms apart at the same posture, so it tests only literal in-sample fitting. TRIAL holds one whole trial out and estimates from the rest. POSTURE removes the offset within quintiles of the reference thigh-shank angle, which is the confound Table 3c identifies: the residual after removing a participant MEAN still carries 2.82-6.19 mm of posture-dependent offset, larger than the effect read out of it.

contrast	joint	in sample	odd/even	trial-out	posture	p trial	p posture
adaptive criterion	hip	+0.264	+0.313	+0.263	+0.322	0.0010	0.0088
adaptive criterion	knee	+0.181	+0.161	+0.181	+0.341	0.0547	0.0088
adaptive criterion	ankle	+0.150	+0.122	+0.149	+0.095	0.1104	0.2793
criterion at matched k=1	hip	-0.718	-0.813	-0.717	-0.781	0.0010	0.0010
criterion at matched k=1	knee	+0.399	+0.412	+0.398	+0.495	0.0039	0.0020
criterion at matched k=1	ankle	+0.076	+0.089	+0.075	+0.067	0.3125	0.3408
criterion at matched k=2	hip	-0.108	-0.193	-0.108	-0.060	0.5752	0.7520
criterion at matched k=2	knee	+0.738	+0.745	+0.736	+0.909	0.0029	0.0020
criterion at matched k=2	ankle	+0.215	+0.202	+0.215	+0.252	0.0684	0.0293
uniform -> confidence	hip	+0.253	+0.216	+0.252	+0.249	0.0098	0.0078
uniform -> confidence	knee	-0.024	-0.017	-0.024	+0.012	0.7285	0.8525
uniform -> confidence	ankle	+0.912	+0.994	+0.912	+0.881	0.0010	0.0010
no rejection -> object-space rejection	hip	-1.726	-1.730	-1.723	-1.924	0.0010	0.0010
no rejection -> object-space rejection	knee	-0.020	-0.023	-0.022	+0.226	0.8320	0.0635
no rejection -> object-space rejection	ankle	+0.256	+0.278	+0.257	+0.261	0.1602	0.1182
no rejection -> reprojection rejection	hip	-1.983	-2.039	-1.979	-2.235	0.0010	0.0010
no rejection -> reprojection rejection	knee	-0.203	-0.188	-0.204	-0.110	0.1992	0.4805
no rejection -> reprojection rejection	ankle	+0.107	+0.155	+0.109	+0.165	0.5967	0.3838

```
=== VERDICT (D), against the rule fixed above ===
  no rejection -> object-space rejection [hip]: -1.726 in sample, -1.723 trial-out (p = 0.0010), -1.924 within posture (p = 0.0010)
      trial-out HOLDS; posture-conditioned HOLDS
  no rejection -> reprojection rejection [hip]: -1.983 in sample, -1.979 trial-out (p = 0.0010), -2.235 within posture (p = 0.0010)
      trial-out HOLDS; posture-conditioned HOLDS
  The hip degradation survives removal of the offset WITHIN posture, so it is not the
  posture-dependent residual Table 3c identifies.

=== (E) IS THE OFFSET-REMOVED EFFECT DISPERSION OR A RESIDUAL MEAN SHIFT? ===

Part A subtracts one vector from both arms, so each arm keeps a residual mean. The effect on
|e - b| is split into the part from those residual means and the part from scatter about
them. POSITIVE = the second-named arm is better, as everywhere else.

contrast                     joint     total  mean-shift  dispersion  disp share
adaptive criterion             hip    +0.264      +0.000      +0.281        1.06
adaptive criterion            knee    +0.181      +0.000      +0.173        0.96
adaptive criterion           ankle    +0.150      +0.000      +0.125        0.83

criterion at matched k=1       hip    -0.718      +0.000      -0.727        1.01
criterion at matched k=1      knee    +0.399      +0.000      +0.338        0.85
criterion at matched k=1     ankle    +0.076      -0.000      +0.041        0.54

criterion at matched k=2       hip    -0.108      +0.000      -0.121        1.13
criterion at matched k=2      knee    +0.738      -0.000      +0.652        0.88
criterion at matched k=2     ankle    +0.215      -0.000      +0.157        0.73

uniform -> confidence          hip    +0.253      +0.000      +0.162        0.64
uniform -> confidence         knee    -0.024      -0.000      -0.016        0.65
uniform -> confidence        ankle    +0.912      +0.000      +1.000        1.10

no rejection -> object-space rejection   hip    -1.726      +0.000      -1.801        1.04
no rejection -> object-space rejection  knee    -0.020      +0.000      -0.125        6.21
no rejection -> object-space rejection ankle    +0.256      +0.000      +0.219        0.86

no rejection -> reprojection rejection   hip    -1.983      +0.000      -2.082        1.05
no rejection -> reprojection rejection  knee    -0.203      +0.000      -0.299        1.47
no rejection -> reprojection rejection ankle    +0.107      +0.000      +0.094        0.88

=== VERDICT (E), against the rule fixed above ===
  no rejection -> object-space rejection: total -1.726, mean-shift +0.000, dispersion -1.801 (dispersion carries 104%)
  no rejection -> reprojection rejection: total -1.983, mean-shift +0.000, dispersion -2.082 (dispersion carries 105%)
  Dispersion carries at least half of each hip effect, so the precision wording in
  Section 3.2 stands.

Section 3.4's verdict pair -- the criterion harmful on {knee, ankle}, beneficial on {hip,
ankle}, null on all three -- is a mean over the same per-joint scalars Part A shows to be
offset-mediated, so the caveat has to be carried into it rather than estimated. Each subset
mean is rebuilt from the same offset-removed observations, per participant, and tested the
same way. Family of four subsets x two retentions, BH within each reference arm (m = 8),

This subsection keeps the two single-arm offsets because Table S4b declares the family that
way; re-declaring it now would be a further degree of freedom. The downward/upward bias
described above pair_scalars applies here too, so the two columns bracket the symmetric
value rather than offering a choice between them.

BH within each reference arm over all 8 tests (4 subsets x 2 retentions).

criterion at matched k=1
  subset              as measured  reference  offset-removed        p        q   verdict
  hip+knee+ankle           +0.022    1st arm          -0.122   0.0479   0.0638   n.s.
  hip+knee+ankle           +0.022    2nd arm          -0.039   0.4307   0.4307   n.s.
  knee+ankle               -0.094    1st arm          +0.218   0.0049   0.0130   SURVIVES
  knee+ankle               -0.094    2nd arm          +0.257   0.0029   0.0059   SURVIVES
  hip+knee                 -0.035    1st arm          -0.214   0.0391   0.0625   n.s.
  hip+knee                 -0.035    2nd arm          -0.103   0.2139   0.2444   n.s.
  hip+ankle                +0.193    1st arm          -0.369   0.0020   0.0130   SURVIVES
  hip+ankle                +0.193    2nd arm          -0.272   0.0029   0.0059   SURVIVES
```

criterion at matched k=2

subset	as measured	reference	offset-removed	p	q	verdict
hip+knee+ankle	+0.031	1st arm	+0.201	0.0391	0.0625	n.s.
hip+knee+ankle	+0.031	2nd arm	+0.361	0.0010	0.0059	SURVIVES
knee+ankle	-0.218	1st arm	+0.412	0.0039	0.0130	SURVIVES
knee+ankle	-0.218	2nd arm	+0.538	0.0020	0.0059	SURVIVES
hip+knee	-0.063	1st arm	+0.211	0.1826	0.2087	n.s.
hip+knee	-0.063	2nd arm	+0.418	0.0098	0.0156	SURVIVES
hip+ankle	+0.374	1st arm	-0.020	0.8076	0.8076	n.s.
hip+ankle	+0.374	2nd arm	+0.128	0.1445	0.1927	n.s.

=== (F) DISPERSION OR CENTROID? THE TEST PART E COULD NOT BE ===

Part E's mean-shift column is zero by construction: it subtracts the midpoint of the two arms' mean errors from both, which leaves residual means of equal norm and opposite sign. It is withdrawn. Below, DISPERSION is scatter about each arm's own centre and CENTROID is the distance of that centre from the reference joint centre; both are free to differ. POSITIVE = the second-named arm is better, as everywhere else.

contrast	joint	total	dispersion	centroid	verdict
adaptive criterion	hip	+0.264	+0.281	-0.310	NOT
adaptive criterion	knee	+0.181	+0.173	+0.291	NOT
adaptive criterion	ankle	+0.150	+0.125	+0.147	NOT
criterion at matched k=1	hip	-0.718	-0.727	+0.554	precision
criterion at matched k=1	knee	+0.399	+0.338	-0.498	NOT
criterion at matched k=1	ankle	+0.076	+0.041	+0.190	NOT
criterion at matched k=2	hip	-0.108	-0.121	+0.744	NOT
criterion at matched k=2	knee	+0.738	+0.652	-1.025	NOT
criterion at matched k=2	ankle	+0.215	+0.157	+0.236	NOT
uniform -> confidence	hip	+0.253	+0.162	+1.676	NOT
uniform -> confidence	knee	-0.024	-0.016	+0.287	NOT
uniform -> confidence	ankle	+0.912	+1.000	-0.498	precision
no rejection -> object-space rejection	hip	-1.726	-1.801	+1.648	precision
no rejection -> object-space rejection	knee	-0.020	-0.125	-0.566	NOT
no rejection -> object-space rejection	ankle	+0.256	+0.219	-0.042	precision
no rejection -> reprojection rejection	hip	-1.983	-2.082	+1.958	precision
no rejection -> reprojection rejection	knee	-0.203	-0.299	-0.856	NOT
no rejection -> reprojection rejection	ankle	+0.107	+0.094	-0.189	NOT

=== VERDICT, against the rule fixed above ===

(1) PRECISION STANDS. At the hip the dispersion contrast carries the sign of the total and at least half its magnitude, and the centroid contrast is the smaller of the two, on both rejection rows. Section 3.2 and the Conclusion may keep the word.

NOTE. Part E is degenerate whichever way this falls, and is reported as such.

=== (G) IS THE REVERSAL A TESTED DIFFERENCE, OR A PAIR OF VERDICTS? ===

Per participant, (offset-removed effect) minus (as-measured effect), same contrast and joint. The offset-removed arm uses the symmetric midpoint, as in Part A. Exact sign-flip over all 2^11 sign patterns; BH within the same declared family of eighteen.

contrast	joint	as measured	offset-removed	difference	exact p	BH q	verdict
adaptive criterion	hip	-0.024	+0.264	+0.288	0.0146	0.0256	SURVIVES
adaptive criterion	knee	+0.339	+0.181	-0.158	0.0332	0.0460	SURVIVES
adaptive criterion	ankle	+0.190	+0.150	-0.040	0.4238	0.4488	n.s.
criterion at matched k=1	hip	+0.253	-0.718	-0.971	0.0010	0.0029	SURVIVES
criterion at matched k=1	knee	-0.322	+0.399	+0.721	0.0020	0.0050	SURVIVES
criterion at matched k=1	ankle	+0.134	+0.076	-0.058	0.4111	0.4488	n.s.
criterion at matched k=2	hip	+0.530	-0.108	-0.638	0.0039	0.0088	SURVIVES
criterion at matched k=2	knee	-0.655	+0.738	+1.393	0.0010	0.0029	SURVIVES
criterion at matched k=2	ankle	+0.219	+0.215	-0.004	0.9639	0.9639	n.s.
uniform -> confidence	hip	+1.564	+0.253	-1.311	0.0010	0.0029	SURVIVES
uniform -> confidence	knee	+0.365	-0.024	-0.389	0.0010	0.0029	SURVIVES
uniform -> confidence	ankle	+0.822	+0.912	+0.090	0.0078	0.0156	SURVIVES
no rejection -> object-space rejection	hip	+0.608	-1.726	-2.335	0.0010	0.0029	SURVIVES
no rejection -> object-space rejection	knee	-0.546	-0.020	+0.526	0.0254	0.0381	SURVIVES
no rejection -> object-space rejection	ankle	+0.426	+0.256	-0.170	0.2402	0.2883	n.s.
no rejection -> reprojection rejection	hip	+0.633	-1.983	-2.616	0.0010	0.0029	SURVIVES
no rejection -> reprojection rejection	knee	-0.885	-0.203	+0.682	0.0156	0.0256	SURVIVES

```
no rejection -> reprojection rejection ankle       +0.237          +0.107      -0.130    0.2051   0.2637   n.s.

=== VERDICT, against the rule fixed above ===
(1) DEMONSTRATED. The difference between the as-measured and offset-removed effects
    survives correction at the hip on both deployable rejection contrasts, so the
    reversal is a tested difference in effect and the title may say 'reverses'.

(3) The matched-k knee rows above are control arms and carry no practical claim either way.
```

Table S8. Exact intervals for the quoted position effects

A sign-flip test at the 0.000977 floor certifies that all eleven participants agreed in direction; it says nothing about magnitude. These are the effects the abstract and Results quote, with the interval obtained by inverting the same exact test -- the procedure already used for the null angle contrasts of Table S3k. Point estimates reproduce the published values exactly, which is what makes the intervals comparable to them.

```
EXACT INTERVALS FOR THE QUOTED POSITION EFFECTS

Exact sign-flip over 11 participants (2^11 = 2048 patterns). The interval is the set of
shifts the same test does not reject at alpha = 0.10, obtained by inverting it -- the
procedure already used for the null angle contrasts of Table S3k, applied here to the
effects the manuscript quotes. POSITIVE = the SECOND-NAMED arm has the lower error.

A subset row is the mean of its member joints, per participant, which is how the subsets are
built throughout; the point estimates reproduce the published values exactly.
```

contrast	effect	exact 95% CI	exact 90% CI	exact p
criterion at matched k=2 [hip]	+0.530	[+0.273, +0.817]	[+0.310, +0.761]	0.0020
criterion at matched k=2 [knee]	-0.655	[-0.939, -0.366]	[-0.882, -0.423]	0.0020
criterion at matched k=2 [ankle]	+0.219	[-0.023, +0.463]	[+0.016, +0.415]	0.0762
criterion at matched k=2 {hip,knee,ankle}	+0.031	[-0.102, +0.175]	[-0.082, +0.145]	0.6543
criterion at matched k=2 {knee,ankle}	-0.218	[-0.360, -0.074]	[-0.334, -0.104]	0.0098
criterion at matched k=2 {hip,ankle}	+0.374	[+0.190, +0.557]	[+0.228, +0.521]	0.0010
criterion at matched k=1 {knee,ankle}	-0.094	[-0.184, -0.003]	[-0.167, -0.019]	0.0459
criterion at matched k=1 {hip,ankle}	+0.193	[+0.020, +0.368]	[+0.053, +0.333]	0.0332
adaptive criterion [hip]	-0.024	[-0.275, +0.227]	[-0.228, +0.186]	0.8350
adaptive criterion [knee]	+0.339	[+0.202, +0.471]	[+0.236, +0.447]	0.0020
adaptive criterion [ankle]	+0.190	[+0.017, +0.360]	[+0.053, +0.326]	0.0352
adaptive criterion [pos:HKA]	+0.168	[+0.064, +0.271]	[+0.084, +0.252]	0.0078
uniform -> confidence [hip]	+1.564	[+1.061, +2.066]	[+1.167, +1.963]	0.0010
uniform -> confidence [knee]	+0.365	[+0.181, +0.566]	[+0.208, +0.526]	0.0010
uniform -> confidence [ankle]	+0.822	[+0.389, +1.307]	[+0.455, +1.208]	0.0010
uniform -> confidence [pos:HKA]	+0.917	[+0.687, +1.175]	[+0.720, +1.118]	0.0010
adaptive reprojection rejection [pos:HKA]	-0.005	[-0.283, +0.249]	[-0.233, +0.224]	0.9639
adaptive object-space rejection [pos:HKA]	+0.163	[-0.064, +0.392]	[-0.023, +0.347]	0.1416
population offset table [pos:HKA]	+9.848	[+7.202, +12.389]	[+7.795, +11.953]	0.0010
population offset table [knee flex]	-0.606	[-0.932, -0.287]	[-0.868, -0.350]	0.0068
GN: reprojection rejection [pos:HKA]	+0.858	[+0.356, +1.390]	[+0.464, +1.262]	0.0020
GN: reprojection rejection [knee flex]	-0.169	[-0.291, -0.045]	[-0.268, -0.071]	0.0146
uniform -> confidence [frontal thigh-shank]	+0.078	[+0.053, +0.104]	[+0.056, +0.099]	0.0010
uniform -> confidence [trunk-thigh]	+0.227	[+0.132, +0.297]	[+0.153, +0.290]	0.0020
adaptive reproj rejection [thigh-shank]	-0.188	[-0.307, -0.066]	[-0.285, -0.089]	0.0098
adaptive reproj rejection [trunk-thigh]	-0.192	[-0.324, -0.060]	[-0.300, -0.087]	0.0107
adaptive objd rejection [thigh-shank]	-0.141	[-0.213, -0.064]	[-0.202, -0.081]	0.0039
adaptive objd rejection [trunk-thigh]	-0.167	[-0.284, -0.044]	[-0.263, -0.068]	0.0146
criterion at matched k=1 [thigh-shank]	-0.089	[-0.154, -0.025]	[-0.140, -0.039]	0.0068
criterion at matched k=2 [thigh-shank]	-0.129	[-0.197, -0.061]	[-0.186, -0.074]	0.0049
population offset table [frontal thigh-shank]	-0.149	[-0.587, +0.297]	[-0.507, +0.199]	0.4434
population offset table [trunk-thigh]	+0.164	[-0.932, +1.205]	[-0.687, +1.022]	0.7314
dose k=1, reprojection [thigh-shank]	-0.286	[-0.478, -0.093]	[-0.441, -0.129]	0.0107
dose k=2, reprojection [thigh-shank]	-0.482	[-0.752, -0.210]	[-0.704, -0.260]	0.0049
dose k=3, reprojection [thigh-shank]	-0.747	[-1.085, -0.408]	[-1.020, -0.477]	0.0010
dose k=1, object-space [thigh-shank]	-0.375	[-0.583, -0.171]	[-0.540, -0.204]	0.0029
dose k=2, object-space [thigh-shank]	-0.612	[-0.913, -0.317]	[-0.852, -0.371]	0.0039
dose k=3, object-space [thigh-shank]	-0.798	[-1.154, -0.445]	[-1.081, -0.512]	0.0020
DLT -> Gauss-Newton [hip]	-2.415	[-3.074, -1.780]	[-2.941, -1.881]	0.0010
DLT -> Gauss-Newton [knee]	-0.781	[-1.190, -0.394]	[-1.116, -0.460]	0.0010
DLT -> Gauss-Newton [ankle]	-0.955	[-1.524, -0.417]	[-1.412, -0.506]	0.0029
GN rejection, w2-matched [hip]	+2.469	[+1.524, +3.417]	[+1.698, +3.241]	0.0010
GN rejection, w2-matched [knee]	-0.616	[-1.019, -0.188]	[-0.949, -0.278]	0.0107
GN rejection, w2-matched [ankle]	+1.316	[+0.632, +2.032]	[+0.720, +1.883]	0.0010
DLT -> Gauss-Newton [pos:HKA]	-1.384	[-1.794, -0.991]	[-1.717, -1.060]	0.0010
GN rejection, w2-matched [pos:HKA]	+1.057	[+0.528, +1.635]	[+0.635, +1.497]	0.0020
GN rejection, w2-matched [knee flex]	-0.175	[-0.309, -0.042]	[-0.281, -0.068]	0.0186

```
HOW TO READ THIS. The 95% interval is the one to read beside a 5%-level verdict, and it is
what the manuscript quotes. The 90% interval is kept because it is the equivalence bound: it
is the one that corresponds to two one-sided tests at 5%, and it is what the null rows are
read against. A 90% interval excluding zero corresponds to p < 0.10, NOT p < 0.05 -- the
matched k=2 ankle row excludes zero at p = 0.076 on the 90% interval and not on the 95%.
Where an interval is wide relative to its point estimate, the effect's SIZE is not
established even though its DIRECTION is, and no recommendation should be scaled to the
point estimate.
```

Table S8 as first built carried no [pos:AW] rows, so the largest triangulation effect in the paper -- confidence weighting against uniform at the ankles and wrists, quoted in Section 5 and Table 1(a) -- was the one headline magnitude with no interval. The rows below repair that. They are computed from the per-participant accumulator that biocv_permutation_v2.py persists, by the same inversion of the exact test, so they need no recomputation of the arms. They also settle a second question: 5.062 mm (Section 5) and 5.086 mm (Table S2c) are the same contrast weighted two ways, the first taking the mean of eleven per-participant differences and the second the difference of frame-pooled means, which weights a participant by how many frames they contribute.

```
EXACT INTERVALS FOR THE [pos:AW] CONTRASTS, FROM THE PERSISTED ACCUMULATOR

Effect = first arm minus second, the order each label names, in millimetres. Intervals are
exact 95%, constructed by inverting the two-sided sign-flip test over all 2^11 = 2048 sign
patterns -- the same construction as Table S8.

contrast                                             n   effect  published        exact 95% interval   width
[pos:AW] uniform vs confidence weighting            11   +5.062     +5.062          [+4.309, +5.930]   1.621
[pos:AW] 1/d alone vs confidence                    11   +4.971     +4.971          [+4.069, +5.897]   1.828
[pos:AW] does 1/d refine confidence?                11   +0.256     +0.256          [+0.058, +0.462]   0.404
[pos:AW] reprojection vs object-space criterion     11   +0.332     +0.332          [+0.202, +0.458]   0.256

=== VERDICT (1)/(2), against the rule fixed in this file's header ===
(1) REPRODUCES: +5.062 against the published +5.062, within 0.01 mm. The exact
    95% interval on the paper's largest triangulation effect is [+4.309, +5.930] mm,
    a width of 1.621 mm, and Table S8 gains its [pos:AW] rows.

=== VERDICT (3): are 5.062 and 5.086 the same contrast weighted two ways? ===
Frame-pooled means: uniform 21.678 mm, confidence 16.592 mm, difference +5.086 mm.
Per-participant mean of differences: +5.062 mm.
ESTABLISHED: the pooled computation returns 5.086, matching the 5.086 of
    Table S2c. The two published figures are the same contrast under two weightings of
    the participants -- equally, and by frame count -- and both documents may say so.

 participant  uniform mm  confidence mm  difference
         P03      30.803         22.776      +8.027
         P04      25.340         20.457      +4.882
         P06      20.516         15.923      +4.593
         P08      20.951         16.935      +4.016
         P09      16.077         12.115      +3.962
         P10      21.907         17.684      +4.222
         P13      18.963         13.451      +5.512
         P16      21.653         15.018      +6.635
         P17      19.994         15.761      +4.233
         P26      19.586         15.114      +4.472
         P28      22.251         17.126      +5.125
```

Table S9. Does the joint-centre offset belong to the detector?

Every other result here rests on one pose estimator. The same 6,240 frames were re-detected with RTMO-m — one-stage, no separate person detector, a different training regime — leaving ground truth, calibration, person association and the keypoint-to-marker mapping untouched, so the two caches differ only in the detector. Both are scored on the plainest arm: no rejection, confidence weights, weighted DLT. The reading below was fixed in the analysis script before the numbers existed. Both caches are distortion-corrected, as every published result is; scoring a corrected cache against a raw one puts the distortion difference into the detector contrast at exactly the near-camera residuals distortion moves most, and doing so inflates RTMO's random residual roughly 2.4-fold and reverses the ankle offset direction. The table below is the corrected comparison; all three tests pass.

What the three detectors do not vary. RTMPose, RTMO and the HALPE-26 model are all from one implementation family, and HALPE-26 extends COCO annotations rather than replacing them with an independently collected corpus, so its indices 0-16 keep COCO-17 semantics. The three are therefore not three independent replications: they vary the architecture and the training data while holding the landmark CONVENTION fixed. That is what makes them a test of the model rather than of the definition, and it is also the reason the persistence result cannot be read as evidence that a differently DEFINED landmark would show the same offset. No control here tests that.

The two detectors agree on where the joint centre is wrong, per participant, at every joint: the cosines run 0.91 to 1.00 at hip and knee and 0.74 to 0.99 at the ankle, and the share of squared error the offset accounts for reproduces the manuscript's pattern, high at hip and knee and roughly half that at the ankle. What this establishes is that the two architectures place the centre in the same wrong place; it does not establish which placement is closer to the marker-based centre, and it says nothing about a detector trained on a different keypoint convention, since both are scored on COCO-17.

```
DOES THE JOINT-CENTRE OFFSET BELONG TO THE DETECTOR OR TO THE KEYPOINT CONVENTION?

The same frames, ground truth, calibration and association rule; only the detector differs.
Both scored on the plainest arm: no rejection, confidence weights, weighted DLT.

    joint            RTMPose (two-stage)           RTMO (one-stage)
               bias mm  random  share      bias mm  random  share
    L-hip         35.3    17.0    78%         36.4    16.4    81%
    R-hip         37.8    15.3    84%         39.7    15.4    86%
   L-knee         25.8    13.7    75%         29.3    14.1    78%
   R-knee         29.2    14.5    80%         33.3    14.6    84%
  L-ankle         10.9    13.6    40%         12.3    15.6    39%
  R-ankle         11.3    12.8    45%         12.3    14.7    41%

DIRECTION: cosine between the two detectors' mean offset vectors, per participant and joint.
    joint  participants  mean cos     min     max
    L-hip            11      0.97    0.91    0.99
    R-hip            11      0.98    0.91    1.00
   L-knee            11      0.99    0.98    1.00
   R-knee            11      0.99    0.97    1.00
  L-ankle            11      0.94    0.80    0.99
  R-ankle            11      0.94    0.74    0.99

=== VERDICTS, against the rule fixed in this file's header ===
(1) SHARE at hip and knee under RTMO: 82%  ->  NOT detector-specific; the claim may be stated for the keypoint convention
```

```
(2) MAGNITUDE ratio RTMO/RTMPose at hip and knee: 1.09x  ->  within a factor of two
(3) DIRECTION, mean cosine at hip and knee: +0.98  ->  same wrong place; between-detector comparison is contaminated the same way
(4) COVERAGE: sufficient (fewest participants at any joint: 11)
```

Table S10. Pipeline arms

Every arm shares the same detections, calibration and candidate cameras; the named component is the only thing that differs. Arms marked DIAGNOSTIC are not deployable as tested and are included to isolate a mechanism.

component	arms
weighting (in a weighted DLT)	uniform; detector confidence c; inverse distance $1/d$ (DIAGNOSTIC: ground-truth distances); confidence × $1/d$, i.e. c/d (DIAGNOSTIC). A c/d^2 variant was null on both outcomes (Table S2c).
rejection rule	none; adaptive — iteratively drop the worst observation while it exceeds median + $3\hat{\sigma}$ ($\hat{\sigma} = 1.4826 \times \mathrm{MAD}$), to a four-camera floor; matched-k (DIAGNOSTIC) — drop exactly k worst-first, so retention is identical across criteria and only the ordering differs.
rejection criterion	reprojection residual, in pixels; object-space residual, in mm (Eq. 1).
joint-centre offset	none; population table — for each participant and joint, the mean error vector of the other ten, taken in world coordinates on the no-rejection, confidence-weighted arm, subtracted from the estimated centre, so the correction is out of sample in participant; own-mean correction (DIAGNOSTIC), reported only as a ceiling (Table 3a).
estimator	weighted DLT; Gauss—Newton refinement of the weighted squared reprojection residual, initialised from the DLT. The two minimise $\Sigma w r^2$ and $\Sigma w^2 r^2$ respectively; contrasts holding that fixed are labelled w^2-matched.

Table S11. Estimator by rejection

Whether discarding is position-neutral under one estimator and not under the other is a comparison between two verdicts, which the inference this paper cites Gelman and Stern (2006) against cannot settle. This is the interaction test that can. Both estimators' arms come from one accumulation, so the per-participant difference is paired rather than assembled across runs, and the component effects reproduce the published values exactly. The reading was fixed in the analysis script before the numbers existed, and the ANGLE row was a predicted NULL: the paper's account says discarding costs the angle under both estimators, so a surviving interaction there would have counted against it.

```
ESTIMATOR-BY-REJECTION INTERACTION: is discarding position-neutral under DLT ONLY?

Rejection effect under each estimator, per participant, then the difference of the two.
POSITIVE component = rejection LOWERS the error. POSITIVE interaction = rejection does
better under Gauss-Newton than under DLT. Same accumulation, so the pairing is exact.

outcome                  n       DLT        GN   interaction          exact 95% CI          exact 90% CI        p        q   verdict
[pos:HKA] position      11    -0.005    +0.858        +0.863   [ +0.576,  +1.152]   [ +0.625,  +1.096]   0.0010   0.0020   SURVIVES
knee flexion            11    -0.188    -0.169        +0.019   [ -0.025,  +0.061]   [ -0.016,  +0.053]   0.3418   0.3418   n.s.

Units: position in mm, angle in deg. The 95% interval is the one to read beside a
5%-level verdict; the 90% interval is the equivalence bound for the null row.
BH within a declared family of two, fixed in this file's header before the numbers existed.

=== VERDICTS, against the rule fixed in this file's header ===
(1) POSITION, q = 0.0020  ->  the estimator changes what discarding does to position; Section 3.2 may state it as a tested difference
(2) ANGLE, q = 0.3418  ->  no evidence the angle cost differs by estimator, as the paper's account predicts
```

Table S12. A bound on the synchronisation residual

The dataset providers corrected a whole-frame offset from an LED pulse and report an uncorrected sub-frame residual without quantifying it; the copy obtained under our data use agreement carries no descriptor from which to quote a figure. A reviewer may reasonably ask whether that residual inflates the RANDOM half of the decomposition and so deflates the 74-81% offset share. It cannot be looked up, but it can be bounded from the reference trajectories themselves, which is what this does. The reading was fixed in the analysis script before the numbers existed; it fired against the paper, and the threshold it used was the wrong comparison. Both the verdict and the reason it is the wrong comparison are printed below, in that order, rather than the rule being replaced after the fact.

The settled position, stated first. The synchronisation residual is bounded at 3.6 mm typical and 7.7 mm worst case at hip and knee on the mean-speed comparison, which is what the manuscript quotes; on the 95th-percentile comparison the same table gives 15.8 mm, which the manuscript also quotes as the wider bound. Against a 27-37 mm offset either bound leaves the offset the dominant term, and the offset share is reported as a bracket (74-81%, and 69-77% if the entire along-track component were timing) rather than a point. The block below records how that position was reached, including a pre-declared verdict that fired on a comparison later found to be the wrong one.

```
AN UPPER BOUND ON THE SUB-FRAME SYNCHRONISATION RESIDUAL, IN MILLIMETRES

108 walking trials, marker-based reference at 200 Hz. This count exceeds the 104
trials analysed elsewhere because it needs only the reference markers, not video: the
trials excluded for missing or unusable video still carry usable reference trajectories,
and a bound on the residual is better estimated on all of them.
A sub-frame residual is less
than one frame period (5.00 ms); the expected magnitude for a residual uniform within one
frame is half of that (2.50 ms). Speed is the per-frame displacement of the reference
marker itself, so no assumption about gait speed enters.

       joint  median spd  p95 speed  typ. bound  worst bound
                    mm/s       mm/s   mm @2.5ms      mm @5ms
    LEFT_HIP      1461.8     2031.3        3.65        10.16
   RIGHT_HIP      1470.1     2027.0        3.68        10.14
   LEFT_KNEE      1400.9     3139.3        3.50        15.70
  RIGHT_KNEE      1395.2     3155.8        3.49        15.78
  LEFT_ANKLE       790.1     4381.9        1.98        21.91
 RIGHT_ANKLE       795.7     4378.0        1.99        21.89

=== READ ===
Largest worst-case contribution at any joint: 21.91 mm (p95 speed x one frame).
Typical contribution at hip and knee: 3.58 mm (median speed x half a frame).
Section 3.1's systematic offset at hip and knee is 27-37 mm and the random residual it is
compared against is 13-17 mm.

AGAINST THE PAPER: the bound is comparable to the random residual, so a material part
of what Section 3.1 calls random error may be synchronisation. The offset SHARE must
be qualified accordingly, and the random half must not be described as triangulation
noise.

It also largely cancels between arms, which share the same frames, and every contrast in
this paper is between arms; the share is the one quantity it can move.

=== THE RULE ABOVE USED THE WRONG THRESHOLD, AND THIS TABLE THE WRONG ATTRIBUTION ===

Two things are wrong with what precedes, and both are left standing rather than replaced.

First, the threshold fixed in this file's header compares a COMPOUNDED worst case -- the
```

95th percentile of joint speed times a FULL frame period, when the whole-frame offset was already corrected -- against an RMS quantity, in linear mm, while the decomposition it bears on is in SQUARED error. That is why it fired.

Second, and more seriously, the arithmetic this file originally offered as a rebuttal placed the residual in the RANDOM half and concluded it 'can only raise the offset share'. That attribution is wrong. Heading varies little WITHIN a participant -- 1.0 to 14.1 deg for ten of the eleven, 53.7 for P26, against 56.0 across all trials (Table S20) -- and the offset is estimated within participant, so a
CONSTANT sub-frame residual displaces every estimate the same way in the world frame and lands in the BIAS term. Table S14 measures the direction and finds the offset is indeed substantially along-track (|cos| = 0.60 at hip and knee). The correction therefore runs the OTHER way: removing a timing component shrinks the numerator of the share, and the share FALLS rather than rises. Table S14 carries that arithmetic, and Section 3.1 now quotes the resulting bracket instead of a point.

What survives from this table is the magnitude bound alone: the residual is about 3.6 mm at hip and knee on the typical figure and under 7.7 mm at worst. What does NOT survive is this file's original claim about which half of the decomposition it enters.

Table S13. A second detector and training corpus, and ankle dorsiflexion

Table S9 changed the detector ARCHITECTURE on the same 17 keypoints. This changes the CONVENTION: HALPE-26 is a 26-keypoint scheme with a different annotation corpus, run on the same frames with ground truth, calibration and association unchanged, and distortion-corrected like every other cache here. It also carries feet, and the BioCV reference carries LTOE/RTOE and HEEL_L/HEEL_R, so ankle dorsiflexion -- absent from the rest of this paper -- becomes measurable on both sides of the comparison. 103 of the 104 trials re-detected.

The reading was fixed in the analysis script before the numbers existed, including the declared sensitivity: the angle is primary on the big toe and repeated on the heel, and where the two foot endpoints disagree in sign NO VERDICT is reported. Both rejection rows fall to that rule. The segment geometry was then measured to test whether a shorter distal segment explained the disagreement; at 1.5x it does not, and the disagreement is left unexplained rather than argued away.

```
HALPE-26: A DIFFERENT MODEL AND TRAINING CORPUS, AND THE SHANK-FOOT ANGLE

The same frames, ground truth, calibration and association rule; only the detector and
its keypoint convention differ. Both caches distortion-corrected.

PART A -- DOES THE OFFSET SURVIVE A CHANGE OF MODEL AND TRAINING CORPUS?
HALPE-26 carries COCO-17's body landmark definitions at indices 0-16, so at hip,
knee and ankle the DEFINITION is unchanged and only the model and corpus differ.

    joint            RTMPose / COCO-17           RTMPose / HALPE-26
               bias mm  random  share      bias mm  random  share
    L-hip        35.3    17.0    78%         35.7    16.6    77%
    R-hip        37.8    15.3    84%         39.3    15.3    84%
   L-knee        25.8    13.7    75%         24.4    13.7    72%
   R-knee        29.2    14.5    80%         27.6    14.8    77%
  L-ankle        10.9    13.6    40%         10.4    13.9    37%
  R-ankle        11.3    12.8    45%         11.4    12.7    46%

DIRECTION: cosine between the two models' mean offset vectors, per participant.
    joint  participants  mean cos     min     max
    L-hip            11      0.97    0.92    0.99
    R-hip            11      0.98    0.96    0.99
   L-knee            11      0.99    0.89    1.00
   R-knee            11      1.00    0.99    1.00
  L-ankle            11      0.99    0.98    1.00
  R-ankle            11      0.99    0.99    1.00

=== PART A VERDICTS, against the rule fixed in this file's header ===
(A1) SHARE at hip and knee under HALPE-26: 77%  ->  not specific to one model or training run; it says nothing about a different landmark definition
(A2) MAGNITUDE ratio HALPE-26/COCO-17 at hip and knee: 0.98x  ->  within a factor of two
(A3) DIRECTION, mean cosine at hip and knee: +0.98  ->  the two models disagree with the reference in the same direction
(A4) ANKLE, mean cosine: +0.99  ->  the ankle offset is stable across model as well as architecture

PART B -- ANKLE DORSIFLEXION, angle(knee, ankle, toe), reference toe = LTOE/RTOE.

Mean absolute error on the plainest arm: 7.524 deg on the toe angle and 25.202 deg
on the heel angle, over 11 participants. Both baselines are given because an
effect on the heel angle can only be read against its own.

EFFECT = mean|e| of the FIRST-named arm minus mean|e| of the SECOND, the convention of
Table S1. So a POSITIVE row means the second-named arm has the lower error: positive on
a rejection row means discarding HELPS this angle, negative means it COSTS, as it does
at the other three angles.

contrast                                     effect deg   exact p     BH q    n   verdict
```

```
none vs adaptive reprojection rejection          -0.522    0.0010   0.0020  11   NO VERDICT: heel disagrees in sign
none vs adaptive object-space rejection          -0.643    0.0010   0.0020  11   NO VERDICT: heel disagrees in sign
uniform vs confidence weighting                  +0.025    0.8018   0.8018  11   n.s.
none vs population offset table                  +0.981    0.0088     --    11   q withheld: leave-one-out rows are not exchangeable (Section 2.5)

Declared family of four, fixed in this file's header before the numbers existed.

DECLARED SENSITIVITY -- the same four contrasts on the heel (HEEL_L/HEEL_R):
contrast                                     effect deg   exact p
none vs adaptive reprojection rejection          +0.651    0.0010
none vs adaptive object-space rejection          +0.578    0.0010
uniform vs confidence weighting                  +0.123    0.3242
none vs population offset table                 +15.014    0.0010

B5 -- IS THE DISSOCIATION JOINT-SPECIFIC WITHIN ONE CONVENTION?

The thigh-shank angle recomputed on THIS cache and these frames, so both angles carry the
same keypoints. POSITIVE = the offset table LOWERS that angle's error.

quantity                              effect          exact 95% CI   exact p
shank-foot angle, deg                 +0.981   [ +0.386,  +1.548]    0.0088
thigh-shank angle, deg                -0.641   [ -0.954,  -0.329]    0.0029
interaction, deg                      +1.622   [ +0.911,  +2.236]    0.0029
interaction, fractional              +41.391   [+21.929, +59.422]    0.0029

n = 11 participants. Baselines: 7.524 deg on the shank-foot angle, 2.620 deg on the thigh-shank angle.
=== B5 VERDICT, against the rule fixed in this file's header ===
    -> WITHHELD. Both arms are the leave-one-participant-out offset table, whose
       differences are not exchangeable, so the exact p above does not apply and no
       dissociation is claimed here. The effects and intervals stand as description
       only (Section 2.5).

IS THE HEEL FIT TO VETO? The foot keypoints' own 3D offset against their reference markers.
    keypoint   bias mm   random   share  participants
   L-big toe      55.7     32.5     74%            11
   R-big toe      54.6     30.4     76%            11
      L-heel      34.1     28.4     59%            11
      R-heel      33.1     27.1     60%            11

heel/toe offset ratio: 0.61x  ->  same order; the heel is a comparable landmark and the declared veto stands

SEGMENT GEOMETRY, because it bears on how much weight the heel row can carry.
An error e perpendicular to the distal segment of a three-point angle moves that angle
by about e/L radians, so a short distal segment amplifies the same positional error.
               segment  mean length mm   deg per mm
      ankle -> big toe           134.7         0.43
         ankle -> heel            87.0         0.66
         knee -> ankle           391.0         0.15

The heel angle is only 1.5x as sensitive to the same positional error as the
toe angle, which is far too small to explain the disagreement on its own. The
BASELINES carry more of it: the heel angle starts from 25.2 deg of mean absolute
error against the toe angle's 7.5, so the offset table's +15.0 deg is a 60% reduction
against the toe row's 13%. That makes the magnitude reconcilable in kind but not in
size, and it does not touch the SIGN REVERSAL on the two rejection rows, which stays
unexplained. The declared rule applies as written: no verdict where the signs differ.
The offset-table row agrees in sign on both endpoints and is reported as a direction
only; its magnitude is not stable across them and must not be quoted as an effect
size.
```

Table S14. Is the offset a synchronisation lag?

Table S12 bounds the synchronisation residual's MAGNITUDE but says nothing about its DIRECTION, and direction is what decides where it lands. Heading varies by 1.0–14.1 degrees within ten of the eleven participants and by 53.7 within the eleventh (Table S20), and the offset is estimated within participant, so a constant sub-frame lag displaces that participant's estimates the same way in the world frame and biases the offset rather than blurring it. The narrower statement this supports, and the one the manuscript makes, is that a timing lag cannot be the SOURCE of the offset; it does not follow that the residual enters only the random half, and it does not.

The test is that a lag is shared by every joint, so displacement = speed x lag must hold with one lag across all six. It does not: hip and knee move at 1.53 and 1.54 m/s yet carry 22.8 and 16.0 mm along the walking direction, and the implied lag spans 9 to 16 ms, every value above the 5 ms frame period a sub-frame residual is bounded by. The reading was fixed in the analysis script before the numbers existed, and its speed premise was wrong on first writing; the original wording and the correction are both kept in that file.

```
IS THE JOINT-CENTRE OFFSET A SYNCHRONISATION LAG?

A lag of tau displaces a joint by (velocity x tau) ALONG the direction of travel, so it must
be collinear with walking direction AND scale with joint speed, tau being shared by all
joints. Both are tested below against the rule fixed in this file's header.

    joint  offset mm  |cos| with  along-track  speed m/s  implied lag
                       walk dir           mm                      ms
    L-hip       35.3        0.59        21.55       1.53        14.35
    R-hip       37.8        0.64        24.06       1.53        16.06
   L-knee       25.8        0.59        15.52       1.54        10.71
   R-knee       29.2        0.57        16.46       1.54        11.16
  L-ankle       10.9        0.74         8.17       0.92         9.25
  R-ankle       11.3        0.65         7.79       0.86         9.16

=== VERDICTS, against the rule fixed in this file's header ===
(1) COLLINEARITY at hip and knee: |cos| = 0.60  ->  consistent with an along-track displacement; test (2) decides
(2) SPEED SCALING. Hip and knee move at 1.53 and 1.54 m/s yet carry 22.80 and 15.99 mm along-track; the implied lag spans 9.2 to 16.1 ms across the six joints.
    ->  A SHARED TIMING LAG IS EXCLUDED as the main source: one tau cannot give two joints of the same speed different displacements, and no single tau fits all six.
(3) IMPLIED LAG at hip and knee: 13.1 ms  ->  far above one frame period (5.0 ms), so not a sub-frame residual

WHAT A SUB-FRAME RESIDUAL COULD CONTRIBUTE, given it is under one frame period:
    L-hip  at most  7.63 mm (speed x one frame), 21.6% of its 35.3 mm offset
    R-hip  at most  7.67 mm (speed x one frame), 20.3% of its 37.8 mm offset
   L-knee  at most  7.68 mm (speed x one frame), 29.8% of its 25.8 mm offset
   R-knee  at most  7.72 mm (speed x one frame), 26.4% of its 29.2 mm offset
  L-ankle  at most  4.61 mm (speed x one frame), 42.2% of its 10.9 mm offset
  R-ankle  at most  4.29 mm (speed x one frame), 38.1% of its 11.3 mm offset

WHAT THIS COSTS THE OFFSET SHARE OF SECTION 3.1, if the along-track part is timing.
Share = |bias|^2 / (|bias|^2 + random^2). Removing a timing component shrinks the numerator.

    joint  bias mm   share  typ. -   share  worst -   share
    L-hip     35.3     81%    3.82     77%     7.63     73%
    R-hip     37.8     86%    3.84     83%     7.67     80%
   L-knee     25.8     78%    3.84     72%     7.68     64%
   R-knee     29.2     80%    3.86     75%     7.72     69%
  L-ankle     10.9     39%    2.30     29%     4.61     18%
  R-ankle     11.3     44%    2.15     34%     4.29     23%


THE SAME ARITHMETIC ON THE ARM THE HEADLINE QUOTES (Table 3a2, object-space rejection):
    joint  bias mm   share  typ. share  worst share
```

L-hip	35.4	74%	71%	69%
R-hip	36.9	80%	78%	75%
L-knee	27.5	79%	76%	73%
R-knee	30.1	81%	79%	77%
L-ankle	11.5	44%	37%	31%
R-ankle	11.5	44%	38%	34%

Headline arm, hip and knee: 74-81% as measured, 69-77% if the entire along-track component were timing.
Offset magnitude at hip and knee: 27-37 mm as measured, 23-33 mm if the entire along-track component were timing.

Worst-case drop at hip and knee: 10.2 percentage points -> the manuscript must report a BRACKET, not a point, and restate the offset magnitude net of the timing term.

READ. The offset IS substantially along-track, so Section 2.1 may not say a timing residual enters only the random half -- a constant residual under a near-constant within-participant heading lands in the bias term. What the tests above establish is that it cannot be the SOURCE of the offset, and the line above bounds how much of the offset it could contribute at all.

Table S15. Statistical specification in full

Section 2.5 states the design; the justifications and the secondary sensitivities are here, moved out of the main text for length and not abridged in the move.

Why Benjamini–Hochberg rather than a stricter procedure. Its proof assumes independent test statistics. Contrasts within a family here share arms, participants and baselines, so the procedure rests on positive regression dependence, which is assumed rather than verified. That is the field's convention, and the arbitrary-dependence alternative of Benjamini and Yekutieli (2001) is reported alongside every verdict rather than in place of it: 107 of 146 survive and the 39 it removes carry a dagger wherever they are quoted (Table S5).

One correction over the union of all fourteen families. Declaring fourteen families rather than one is a presentational choice, so the pooled alternative is reported too. Over the union ($m = 244$) the count falls from 144 to 141: five verdicts change, one gaining and four losing, and none is a headline claim. Two further contrasts carry no verdict at all, withheld by an endpoint veto declared with their outcome (Table S4).

What a percentage-point interaction figure is. A position-against-angle interaction is the difference of two fractional changes. Its magnitude in percentage points is the tested statistic's effect size on the relative scale, not a physical quantity, and it is not convertible to millimetres or degrees.

Intervals. Exact intervals invert the sign-flip test at 95% where a magnitude is quoted and at 90% where a null is bounded, 90% being the level corresponding to two one-sided 5% tests. Percentile bootstrap intervals use $B = 3000$ and resample participants; where the two disagree the exact test governs (Tables S3k, S8).

Table S16. Dataset, rig and calibration detail

Participants. The eleven present in our copy are P03, P04, P06, P08, P09, P10, P13, P16, P17, P26 and P28. We hold nothing on the other four of the archive's fifteen, so the inferential units are a non-random subset.

Frame sampling. Sixty frames per trial at near-even spacing is a compute bound, not a property of the data; the spacing spans the whole pass through the capture volume rather than concentrating on part of it. No temporal filter can be applied at 45 ms between retained frames, which is why every markerless estimate here is unfiltered.

Two foot-offset figures. Table S13 gives the toe keypoint's offset as 55.7 and 54.6 mm per side and Table S19 as 52.24 mm. They are the same quantity on different frames: S13 requires only that keypoint to be valid, while S19 requires knee, ankle, toe and heel to be simultaneously valid and both endpoints triangulable, because it decomposes each offset against a segment those four points define. The stricter subset is the smaller figure; both are correct for the frames they are computed on.

Keypoint-to-reference correspondence. Each COCO-17 landmark is differenced against the identically named point in the archive's own output: LEFT_SHO, RIGHT_SHO, LEFT_ELBOW, RIGHT_ELBOW, LEFT_WRIST, RIGHT_WRIST, LEFT_HIP, RIGHT_HIP, LEFT_KNEE, RIGHT_KNEE, LEFT_ANKLE and RIGHT_ANKLE. These are joint centres produced by the dataset's 6-DoF model, not raw markers, so every offset in this paper is keypoint minus model joint centre. Under HALPE-26 the two foot endpoints are differenced against the reference markers LTOE/RTOE and HEEL_L/HEEL_R. The archive record states that joint centres are the midpoint of the medial and lateral markers at every joint except the hip, which uses the regression of Bell et al. (1989); that regression carries its own error, of the order of the offset reported here and mirrored across the midline, which is why the hip offset cannot be apportioned between the two systems.

Person association and admission. Per camera, the detected person whose visible COCO joints best match the projected reference is selected, and the camera is dropped for that frame if the best match's median skeleton distance exceeds 80 px. A 2D observation is admitted if its coordinates are finite and its detector score is strictly positive, and a joint is triangulated only where at least four cameras are admitted.

Rig. The nine machine-vision cameras have non-uniform focal lengths and substantial radial distortion across all 99 calibrations; the geometry is specified in Table S0. We used the dataset's updated calibration throughout.

Cross-system alignment. The providers report 3.62 mm on annotated WALK_01 frames and 2.31 mm above the force plates (Evans et al., 2024). Both are their figures; we did not recompute either and no analysis here derives one.

Four thigh-shank baselines. The same angle is quoted at four values because four different arms and frame sets produce it, and Section 2.4 points here for the reconciliation. 2.575 deg is the per-participant mean on the full pipeline arm (Table 3a, Table S2c). 2.570 deg is the same quantity pooled over frames rather than participants (Table 2b). 2.669 deg is the per-participant $k = 0$ baseline of the threshold sweep, which runs on the sweep's own frame set (Table S21).

2.722 deg is the object-space arm with confidence weighting on that same frame set (Tables S21, S3f). None is comparable in absolute value with model-based reporting, and no contrast in this paper is taken across two of them.

Table S17. Why does the population offset table harm the knee angle?

Two mechanisms were tested. Neither is established, and the harm is reported as bounded rather than explained.

Geometry. Of the knee's offset relative to the hip–ankle chord, only the in-plane perpendicular component acts in the direction that moves flexion: 11.04 mm against 27.45 mm axial and 8.63 mm out of plane (Table 3b). This cannot explain the harm, because removing the ENTIRE true offset changes the angle by 0.166°, less than the 0.606° the table costs.

Mismatch. The difference between a participant's own offset and the population mean is of ample size: applied to the reference points and the angle recomputed exactly, it predicts 2.19–2.33° against the 0.606° observed. But the harm does not track it across participants ($r = -0.40$ to $+0.31$, p at least 0.23). With $n = 11$ the test cannot detect a correlation below about 0.6, so this is an absence of evidence at low power, not evidence against the mechanism (Table S3e).

Table S18. Does the absence of temporal filtering change the result?

Every error in the main analysis is an unfiltered markerless estimate against a 12 Hz-filtered reference, because trials are sampled every ninth frame and 45 ms between retained frames admits no temporal filter. The sampling was a compute choice and not a property of the data, so the measurement is available: contiguous windows were re-detected for a subset of trials and the reference's own filter applied to the markerless trajectories. The reading below was fixed in the analysis script before the numbers existed, including the condition under which the angle conclusion would have to be withdrawn.

DOES FILTERING MOVE THE OFFSET SHARE?

44 trials from 11 participant(s), contiguous windows of up to 600 frames.
Markerless trajectories filtered with the reference's own filter: Butterworth, order 4, 12 Hz, zero-phase.

joint	bias raw	bias filt	rand raw	rand filt	share raw	share filt	shift
L-hip	35.5	35.5	16.9	16.6	78.9%	79.5%	+0.54
R-hip	37.1	37.2	15.6	15.3	83.0%	83.5%	+0.49
L-knee	25.4	25.4	14.0	13.6	73.5%	74.4%	+0.92
R-knee	28.8	28.8	15.2	14.8	77.9%	78.6%	+0.70
L-ankle	10.9	10.9	13.1	12.6	43.4%	44.7%	+1.29
R-ankle	11.0	11.0	12.3	11.8	46.2%	47.6%	+1.38

=== VERDICTS, against the rule fixed in this file's header ===
(1) DIRECTION: filtering moves the hip/knee share by +0.7 points -> the manuscript's stated direction is CONFIRMED and may be quoted as measured
(2) SIZE: 0.7 points -> under one point; the sampling choice did not matter and the caveat can be retired
(3) SCOPE: a subset, so this settles the direction and bounds the size on these participants. It does not replace the headline figure.

DOES THE REJECTION COST TO THE ANGLE SURVIVE FILTERING?

Thigh-shank angle, mean absolute error in degrees against the filtered reference.
COST = rejection minus no rejection; positive means rejection is worse, which is the direction the manuscript reports. Filtering is applied to the joint POSITIONS and the angle formed afterwards.

pid	none raw	objd raw	cost raw	none filt	objd filt	cost filt
P03	1.950	2.191	+0.240	1.907	2.080	+0.174
P04	2.845	2.903	+0.058	2.816	2.838	+0.023
P06	3.119	3.315	+0.196	3.114	3.293	+0.179
P08	2.698	2.776	+0.078	2.660	2.673	+0.013
P09	2.010	2.166	+0.156	1.974	2.053	+0.079
P10	2.542	2.449	-0.093	2.527	2.359	-0.168
P13	2.741	2.948	+0.207	2.718	2.851	+0.133
P16	2.243	2.585	+0.342	2.206	2.499	+0.294
P17	1.984	2.218	+0.235	1.964	2.060	+0.096
P26	5.279	5.328	+0.049	5.286	5.348	+0.062
P28	1.861	1.889	+0.028	1.841	1.758	-0.082

Mean cost raw +0.136 deg, filtered +0.073 deg (11 participant(s)).

=== VERDICT (5), against the rule fixed in this file's header ===
HOLDS (0.073 against 0.136, at least half). The angle conclusion stands and may be quoted as robust to the absence of temporal filtering, on this subset.

SCOPE: a subset of participants and trials. It settles the DIRECTION and bounds the size of the filtering bias on the angle; it does not replace the headline figure.

=== POST HOC: is the filtered cost distinguishable from zero? ===
Declared post hoc. This test was added after the per-participant numbers were seen; it is not part of the rule fixed in this file's header, and it is not carried into any declared family.
raw: mean +0.136 deg, 1 of 11 participants negative, exact sign-flip p = 0.0078
filtered: mean +0.073 deg, 2 of 11 participants negative, exact sign-flip p = 0.0840

Table S19. Does geometry explain the toe/heel disagreement?

Table S13 leaves the endpoint disagreement unexplained and rejects a sensitivity account on the ground that the heel is only 1.5 times as sensitive per millimetre. That compares the wrong quantity: what moves a three-point angle is the offset component PERPENDICULAR to the segment, in the plane of the angle, not the offset's length. The decomposition below is the one already run for the knee in Table 3(b), applied to each foot endpoint against its own ankle-to-endpoint segment. Its reading was fixed before it was run, including the branch under which the result would count against the paper's choice of primary endpoint.

```
WHY THE TOE AND HEEL ENDPOINTS DISAGREE: THE OFFSET'S DIRECTION, NOT ITS LENGTH

Each endpoint's mean offset is decomposed against ITS OWN ankle-to-endpoint segment, in the
plane of angle(knee, ankle, endpoint). Only the in-plane perpendicular component moves that
angle. Baselines are Table S13's mean absolute angle errors: toe 7.524 deg, heel 25.202 deg.

 endpoint    n  |offset|    axial  in-plane  out-plane  |segment|  implied deg  of baseline
      toe   11     52.24    47.31     21.51       2.10      134.4         8.99       119.5%
     heel   11     33.02    11.79     29.49       3.68       86.6        19.10        75.8%

=== VERDICT, against the rule fixed in this file's header ===
Toe:  the offset implies 8.99 deg of angle error, 119.5% of its 7.524 deg baseline.
Heel: the offset implies 19.10 deg of angle error, 75.8% of its 25.202 deg baseline.

(2) NOT EXPLAINED: the two implied angle offsets are comparable fractions of their
    baselines, so geometry does not account for the disagreement. Table S13's
    statement stands unchanged and this file adds only a bound.

(4) The veto stands either way: Table S13 returns no verdict where the two endpoints
    disagree in sign, and explaining a disagreement is not agreement.
```

Table S20. The span of walking headings

Three claims rest on how widely heading varies: the non-transferability of the world-frame offset table, the near-identity of the world- and body-frame decompositions in Table S3m, and the argument in Table S14 that a constant sub-frame timing residual lands in the bias term. Heading is measured here from the reference alone. It varies widely ACROSS trials and narrowly WITHIN a participant, which is the level at which the offset is estimated. The reading below was fixed before the number existed.

```
THE SPAN OF WALKING HEADINGS, MEASURED

Heading is the horizontal direction of mid-hip displacement from the first to the last
usable reference frame of a trial. Opposite directions of travel along one axis are the same
line, so the arc is computed on the doubled angle and halved.

109 trials from 11 participants.
Table S12 counts 108 for a reference-only quantity because it requires all six
lower-limb joints to be labelled, where heading needs only the two hips; the extra
trial has usable hips and an unusable joint elsewhere.
Smallest arc containing every heading, as an axis: 56.0 degrees.

 participant  trials  own span, deg
         P03      10            7.7
         P04       9            3.0
         P06      10           14.1
         P08      10            1.5
         P09      10            1.0
         P10      10            2.9
         P13      10            2.2
         P16      10            1.5
         P17      10            1.3
         P26      10           53.7
         P28      10            1.0

=== VERDICT, against the rule fixed in this file's header ===
(2) WIDER THAN ASSERTED: 56.0 degrees. The prose figure of 9 degrees is wrong.
    The world/body near-identity must be justified from Table S3m directly, and the
    timing argument restated as applying to the heading-common component only.
```

Table S21. Is the adaptive rule's threshold load-bearing?

Section 2.2 fixes the adaptive rule at a median-plus-three-MAD threshold with a four-camera floor. Both numbers were chosen and, until this table, never varied, so every rejection result in the paper was conditional on them. The multiplier is swept here over 2.5, 3.0, 3.5 and 4.0 on both criteria and the same frames, with the reading fixed before the numbers existed, including the branch under which the angle cost would have had to be restated.

```
IS THE ADAPTIVE RULE'S THRESHOLD LOAD-BEARING?

Both criteria at four MAD multipliers against no rejection, on the same frames.
POSITIVE effect = rejection is better. Per participant, then averaged.

     arm   kept/9  [pos:HKA] mm  effect mm   thigh-shank deg  effect deg
    none     8.92        28.280     +0.000            2.669      +0.000
objd k=2.5     8.22        28.172     +0.108            2.889      -0.219
objd k=3.0     8.45        28.117     +0.163            2.810      -0.141
objd k=3.5     8.59        28.098     +0.182            2.757      -0.088
objd k=4.0     8.68        28.081     +0.199            2.722      -0.053
rep  k=2.5     8.00        28.371     -0.092            2.955      -0.286
rep  k=3.0     8.26        28.285     -0.005            2.857      -0.188
rep  k=3.5     8.43        28.226     +0.054            2.809      -0.140
rep  k=4.0     8.54        28.184     +0.096            2.776      -0.107

=== VERDICT, against the rule fixed in this file's header ===
(1) POSITION: the largest effect across multipliers is 0.199 mm.
    Every multiplier keeps the contrast inside the +/-0.35 mm bound Section 3.2 quotes,
    so that bound is not an artefact of the threshold.
(2) ANGLE: effects objd k=2.5: -0.219, objd k=3.0: -0.141, objd k=3.5: -0.088, objd k=4.0: -0.053, rep k=2.5: -0.286, rep k=3.0: -0.188, rep k=3.5: -0.140, rep k=4.0: -0.107 deg.
    The sign holds but the magnitude varies by more than a factor of two, so the
    quoted size is conditional on the multiplier even though its direction is not.
(3) Retention falls monotonically with the multiplier, which is the dose effect the
    dose-response family already reports; it is not itself a verdict here.
```

Table S22. What the withheld leave-one-out rows did to everyone else's q

Withholding a q is not the whole consequence of an invalid p-value. Those p-values were part of the Benjamini-Hochberg ranking of the families they sat in, so every other q in those families was computed against them. Each affected family is corrected again without the leave-one-out rows, with the reading fixed before the numbers existed.

```
WHAT THE WITHHELD LEAVE-ONE-OUT ROWS DID TO EVERYONE ELSE'S q

Benjamini-Hochberg is recomputed within each declared family with the leave-one-out rows
removed. Only rows that REMAIN are compared; the removed rows have no q by the rule of
Section 2.5.

Survivor counts are over the ROWS THAT REMAIN, so 'with' and 'without' differ only
through the ranking the removed rows imposed. Table captions that quote a survivor count
out of the DECLARED family size are quoting a denominator that includes rows with no q.

family                           m  m without LOO  surv with  surv without  verdicts moved
main contrasts                  50             46         25            25               0
position vs angle               20             17         10            10               0
within-limb: membership, %      30             24         15            15               0
within-limb: membership, mm     30             24         10            10               0
within-limb: subset alone       20             16         10             9               1

=== VERDICT (2), against the rule fixed in this file's header ===
1 verdict(s) move when the leave-one-out rows are removed:

family                      contrast                                          p   q with  q without   change
within-limb: subset alone    criterion at matched k=1                     0.0332   0.0474     0.0531   loses

Each of these is published on the reduced family, the sizes are those Table S4b
states, and the survivor count Section 2.5 reports is the reduced one.
```

Table S23. Is the triangulation null a property of the rule, or of a redundant rig?

Every rejection result in the manuscript is measured on a nine-camera ring where the adaptive rule retains 8.26-8.68 cameras of nine (Table S21). Under that redundancy discarding one observation is close to a no-op, so a small effect is what the geometry predicts rather than what the pipeline reveals, and most laboratories the recommendation would reach run four to eight cameras. The cached detections hold all nine views, so the same contrasts are recomputed on fixed camera subsets at no new detection cost. The subsets are chosen by index to spread around the ring and are ONE choice, not an average over all subsets of that size; the four-camera rejection floor is unchanged. The reading, including the branch under which the manuscript would have to narrow its headline claim, was fixed in the script before the numbers existed. That branch is the one it returned.

```
IS THE TRIANGULATION NULL A PROPERTY OF THE RULE, OR OF A REDUNDANT RIG?

Same frames, same detections, same adaptive rule at k = 3 MAD with a four-camera floor.
Only the set of candidate cameras changes. EFFECT = no rejection minus rejection, per
participant then averaged; POSITIVE means rejection is better. Frames are those where all
six lower-limb joints keep at least four cameras within the subset, so the frame set
narrows as cameras are removed and the LEVELS are not comparable across rows -- the
CONTRAST is.

     subset    arm   kept  position mm  effect mm   exact p  angle deg  effect deg   exact p
  9 cameras   none   8.92       28.280     +0.000       nan      2.669      +0.000       nan
  9 cameras    rep   8.26       28.285     -0.005    0.9639      2.857      -0.188    0.0098
  9 cameras   objd   8.45       28.117     +0.163    0.1416      2.810      -0.141    0.0039
  7 cameras   none   6.92       28.612     +0.000       nan      2.922      +0.000       nan
  7 cameras    rep   6.41       28.553     +0.059    0.6533      3.129      -0.206    0.0059
  7 cameras   objd   6.55       28.448     +0.164    0.1357      3.068      -0.145    0.0068
  5 cameras   none   4.99       30.045     +0.000       nan      2.886      +0.000       nan
  5 cameras    rep   4.72       29.188     +0.857    0.0010      3.006      -0.120    0.0225
  5 cameras   objd   4.79       29.616     +0.428    0.0029      2.931      -0.045    0.1592

=== VERDICTS, against the rule fixed in this file's header ===
(2) AGAINST THE PAPER: the null IS rig-dependent. 2 contrast(s) leave the
    +/-0.35 mm bound once cameras are removed:
      5 cameras, rep: +0.857 mm
      5 cameras, objd: +0.428 mm
    The null of Sections 3.2 and 5 is therefore conditional on a nine-camera ring,
    and a sparser rig was not tested to the same conclusion.

(3) ANGLE: the cost of discarding keeps its sign at every camera count, so its direction
    is not an artefact of redundancy.

(5) One subset per size, not an average over subsets. This is evidence about these camera
    sets, not about five-camera rigs in general.
```